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Preface

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List of Notations

The following table lists the notation used throughout this report, organized by topic.

Symbol	Meaning
General	
$\mathbb{N}$	Set of all natural numbers i.e. positive integers
$\mathbb{Z}$	Set of all integers
$\mathbb{R}$	Set of real numbers
$\mathbb{C}$	Set of complex numbers
S^1	$\{z \in \mathbb{C} : \lvert z\rvert = 1\}$. The multiplicative group of complex numbers of absolute value 1
ζ_m	A primitive m-th root of unity
Finite Fields	
$\mathbb{F}_q$	Finite field with q elements
$\mathbb{F}_q^*$	Multiplicative group of nonzero elements of $\mathbb{F}_q$
$\mathbb{F}_p$	Prime field with p elements
$(\mathbb{F}_q^*)^{(d)}$	Subgroup of d-th powers in $\mathbb{F}_q^*$
$[E : \mathbb{F}_q]$	Degree of the extension $E/\mathbb{F}_q$
E, F, L, K	Generic notation for finite fields or finite extensions of finite fields
Characters of Finite Abelian Groups	
G	A finite abelian group with identity element 1_G
Z_k	Cyclic multiplicative group of order k
χ_0	Trivial character of G: $\chi_0(g) = 1$ for all $g \in G$
$\overline{\chi}$	Conjugate (and inverse) character of χ: $\overline{\chi}(g) = \overline{\chi(g)}$
$\mathrm{Hom}(A, B)$	Set of all group homomorphisms from A to B
$\widehat{G} = \mathrm{Hom}(G, S^1)$	Dual group of G; the group of all characters of G
$\mathrm{Ann}(\chi)$	Annihilator of character χ: $\{g \in G : \chi(g) = 1\}$, a subgroup of G
$\mathrm{Ann}(H)$	Annihilator of subgroup $H \leq G$: $\{\chi \in \widehat{G} : \chi(h) = 1\ \forall h \in H\}$, a subgroup of $\widehat{G}$
ev_g	Evaluation character of $\widehat{G}$ defined by $\mathrm{ev}_g(\chi) = \chi(g)$
Maps on Finite Fields	
$\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$	Absolute trace function
$\mathrm{Tr}_{E/\mathbb{F}_q}$	Trace function from E to $\mathbb{F}_q$

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Symbol	**Meaning**
$\mathrm{N}_{E/\mathbb{F}_q}$	Norm function from E to $\mathbb{F}_q$
$\sigma : \mathbb{F}_q \to \mathbb{F}_q$	Frobenius automorphism: $\sigma(\alpha) = \alpha^p$
Additive Characters of $\mathbb{F}_q$	
χ, τ	Generic notation for additive characters
χ_1	Canonical additive character of $\mathbb{F}_q$: $\chi_1(\alpha) = e^{2\pi i \,\mathrm{Tr}(\alpha)/p}$
χ_β	Additive character defined by $\chi_\beta(\alpha) = \chi_1(\beta\alpha)$, for $\beta \in \mathbb{F}_q$
$\boldsymbol{\mathcal{X}}_q$	Set of all additive characters of $\mathbb{F}_q$
$\chi^{(r)}$	Additive character of $\mathbb{F}_q$ lifted to E by $\chi^{(r)} = \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$ with $[E : \mathbb{F}_q] = r$
Multiplicative Characters of $\mathbb{F}_q$	
ψ, λ	Generic notation for multiplicative characters
ψ_0	Trivial multiplicative character of $\mathbb{F}_q$
ψ_j	Multiplicative character defined by $\psi_j(g^k) = e^{2\pi i jk/(q-1)}$, for $0 \le j \le q-2$
η	Quadratic character of $\mathbb{F}_q$ (for odd q): $\eta(\alpha) = 1$ if α is a square in $\mathbb{F}_q^*$, and -1 otherwise
$\boldsymbol{\mathcal{M}}_q$	Set of all multiplicative characters of $\mathbb{F}_q^*$
$\boldsymbol{\mathcal{M}}_q^{(e)}$	Set of multiplicative characters of $\mathbb{F}_q$ of exponent e
$\mathrm{ord}(\psi)$	Order of ψ: smallest positive integer d with $\psi^d = \psi_0$
$\left(\frac{\alpha}{q}\right)$	Legendre symbol; equals $\eta(\alpha)$ for $\alpha \in \mathbb{F}_q$
$\psi^{(r)}$	Multiplicative character of $\mathbb{F}_q$ lifted to E by $\psi^{(r)} = \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$ with $[E : \mathbb{F}_q] = r$
Exponential Sums	
$G(\psi, \chi)$	Gaussian sum: $\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha)\,\chi(\alpha)$, for $\psi \in \boldsymbol{\mathcal{M}}_q$, $\chi \in \boldsymbol{\mathcal{X}}_q$
$J_\alpha(\lambda_1, \ldots, \lambda_k)$	Generalized Jacobi sum: $\sum_{\substack{\alpha_1,\ldots,\alpha_k \in \mathbb{F}_q \\ \alpha_1+\cdots+\alpha_k=\alpha}} \lambda_1(\alpha_1)\cdots\lambda_k(\alpha_k)$, for $\lambda_i \in \boldsymbol{\mathcal{M}}_q$
$J(\lambda_1, \ldots, \lambda_k)$	Jacobi sum; the special case $\alpha = 1$ of J_α
$J_0(\lambda_1, \ldots, \lambda_k)$	Jacobi sum at 0; the special case $\alpha = 0$ of J_α
Dirichlet L-functions over $\mathbb{F}_q[X]$	
$\zeta(s)$	Riemann-Zeta function for $s \in \mathbb{C}$
$L(s, \chi)$	Dirichlet L-function: $\sum_{h \in \Phi} \chi(h)\,\mathfrak{n}(h)^{-s}$, for a multiplicative character χ
$\overline{L}(z, \chi)$	Power series associated to $L(s, \chi) = \sum_{h \in \Phi} \chi(h)\, z^{\deg(h)}$, satisfying $\overline{L}(q^{-s}, \chi) = L(s, \chi)$
Φ	Set of all monic polynomials in $\mathbb{F}_q[X]$
$\mathrm{Irr}(\Phi)$	Set of irreducible monic polynomials in Φ
L_k	k-th coefficient of $z \frac{d}{dz} \log \overline{L}(z, \chi) = \sum_{h \in \mathrm{Irr}(\Phi),\, \deg(h) \mid k} \deg(h)\, \chi(h)^{k/\deg(h)}$

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Chapter 1

Introduction

Chapter 2

Some Basics of Finite Fields

Chapter 3

Exponential Sums over Finite Fields

Exponential sums are important tools in number theory for solving problems involving integers. These sums have many applications in Analytic Number Theory.

In this chapter we will set up exponential sums for finite fields by special group homomorphisms called *characters*. Since a finite field has two different inherent group structure; one is additive and the other is multiplicative we obtain two different characters. Exponential sums are formed by using the values of one or more characters and possibly combining them with weights or with other function values.

3.1 Characters of Finite Abelian Groups

Let G be a finite abelian group with the identity element 1_G. Let S^1 be the set of complex numbers with absolute value 1. Then consider the multiplicative group S^1 with the operation of multiplication.

Definition 3.1.1: Character χ of G

A character χ of G is a group homomorphism $\chi : G \longrightarrow S^1$ to the multiplicative group S^1 i.e. for all $g_1, g_2 \in G$

$$\chi(g_1 \cdot g_2) = \chi(g_1) \cdot \chi(g_2)$$

So if χ is any character of G then $\chi(1_G) = 1$. Since by definition for all $g \in G$,

$$\chi(g)^{|G|} = \chi(g^{|G|}) = \chi(1_G) = 1$$

the values of χ are $|G|^{th}$ roots of unity. Also we have for any $g \in G$,

$$\chi(g)\chi(g^{-1}) = \chi(g \cdot g^{-1}) = \chi(1_G) = 1$$

so $\chi(g^{-1}) = \chi(g)^{-1} = \overline{\chi(g)}$.

There are some special characters of G which have specific name. The character χ_0 defined by $\chi_0(g) = 1$ for all $g \in G$ is called the *trivial character*. For any character χ of G, the character $\overline{\chi}$ defined by $\overline{\chi}(g) = \overline{\chi(g)}$ for all $g \in G$ is called the *conjugate character* of χ.

Given any finitely many characters of G, $\chi_1, \ldots, \chi_n$ we can define the product character $\chi = \chi_1 \cdot \chi_2 \cdots \chi_n$ such that for all $g \in G$,

$$\chi(g) = \chi_1(g) \cdot \chi_2(g) \cdots \chi_n(g)$$

Hence for all $g \in G$,

$$\chi(g) \cdot \overline{\chi}(g) = |\chi(g)|^2 = 1$$

Hence the $\overline{\chi}$ is also the inverse character of χ. Hence the set of all characters of G forms a group, $\mathrm{Hom}(G, S^1)$ under multiplication whose identity element is the trivial character χ_0.

Definition 3.1.2: Dual Group to G

For a finite abelian group G, the group of all characters of G with the identity element χ_0 is called the dual group *to G and is denoted by $\widehat{G}$ i.e. $\widehat{G} := \mathrm{Hom}(G, S^1)$.*

Observation 3.1. *The dual group $\widehat{G}$ is a finite abelian group. So we can talk about characters of $\widehat{G}$ too.*

Since the values of characters of G can only be $|G|^{th}$ roots of unity, the dual group $\widehat{G}$ is a finite abelian group.

Lemma 3.1.1

If G is a finite cyclic group of order n and let g be the generator of G, then for all $j \in \{0, 1, \ldots, n-1\}$ define the function

$$\chi_j(g^k) = e^{2\pi i \frac{j \cdot k}{n}} \quad \forall\, k \in \{0, 1, \ldots, n-1\}$$

Then $\widehat{G} = \{\chi_j : j \in \{0, 1, \ldots, n-1\}$.

Proof: Certainly for all $j \in \{0, 1, \ldots, n-1\}$ defines a character of G. On the other hand suppose, $\chi \in \widehat{G}$. Then $\chi(g)$ must be an nth root of unity. Hence there exists $j \in \{0, 1, \ldots, n-1\}$ such that $\chi(g) = e(j/n)$. Therefore $\chi = \chi_j$. Hence $\widehat{G} = \{\chi_j : 0 \le j \le n-1\}$. ■

Observation 3.2. *The characters defined in above lemma forms a cyclic group with χ_1 being the generator of $\widehat{G}$.*

Remark: When $G = (\mathbb{Z}/n\mathbb{Z})^*$, the characters of G described by $\chi_j(\alpha) = e^{2\pi i \frac{\alpha \cdot j}{n}}$ like in Lemma 3.1.1 are precisely the classical *Dirichlet characters* modulo n, which are of fundamental importance in analytic number theory.

Corollary 3.1.2

If G is a finite cyclic group of order n with g being the generator then $G \cong \widehat{G}$ by the isomorphism $\varphi : g^j \mapsto \chi_j$ for all $j \in \{0, 1, \ldots, n-1\}$.

Now we will show that we can extend a character of any subgroup to a character of the group.

Theorem 3.1.3

Let H be a subgroup of the finite abelian group G and let χ_H be a character of H. Then χ_H can be extended to a character of G that is there exists a character $\chi \in \widehat{G}$ with $\chi(h) = \chi_H(h)$ for all $h \in H$.

Proof: Let H be a proper subgroup of G. Let $\alpha \in G \setminus H$. Then let H_1 be the group generated by H and α i.e. $H_1 = \langle H, a \rangle$. Let m be the smallest positive integer such that $\alpha^m \in H$. Then for all $g \in H_1$, g can be written uniquely as $g = \alpha^j \cdot h$ where $0 \le j < m$ and $h \in H$.

Let ω be a complex number such that $\omega^m = \chi_H(\alpha^m)$. Then for all $g \in H_1$ where $g = \alpha^j \cdot h$, $h \in H$ define the function $\chi_1(g) = \omega^j \chi_H(h)$. Then for any $g_1, g_2 \in H_1$ where $g_1 = \alpha^i \cdot h_1$, $g_2 = \alpha^j \cdot h_2$, we have $g_1 \cdot g_2 = \alpha^{j+k} \cdot h_1 \cdot h_2 = \alpha^{j+k \bmod m} \cdot h_1 \cdot h_2$. Hence

$$\chi_1(g_1 \cdot \chi_2) = \omega^{j+k \bmod m} \chi_H(h_1) \cdot \chi_H(h_2) = \alpha^j \chi_H(h_1) \cdot \omega^k \chi_H(h_2) = \chi_1(g_1) \cdot \chi_1(g_2)$$

Hence χ_1 is a character of H_1. And also for all $h \in H$ we have $\chi_1(h) = \chi_H(h)$.

So if $H_1 = G$ we are done. Otherwise we can continue this process above until we get G. This process we have to do finitely many times as G is a finite group. ■

Therefore if we have two distinct elements $g_1, g_2 \in G$, $g_1 \neq g_2$ then take the subgroup generated by the element $h = g_1 \cdot g_2^{-1} \neq 1$. Since this subgroup is a cyclic subgroup generated by h, by Lemma 3.1.1 we get a character χ' of the cyclic subgroup such that $\chi'(h) \neq 1$. Then by Theorem 3.1.3 we get a character χ of G such that $\chi(h) = \chi'(h) \neq 1$ and hence $\chi(g_1) \neq \chi(g_2)$. Therefore we have the following observation

Observation 3.3. *For any two distinct elements $g_1, g_2 \in G$, $g_1 \neq g_2$ we can get a character $\chi \in \widehat{G}$ such that $\chi(g_1) \neq \chi(g_2)$.*

Now consider any character χ of G. If $g \in G$ such that $\chi(g) = 1$ then $\chi(g^{-1}) = 1$. Hence the set of all elements $g \in G$ such that $\chi(g)$ forms a group. This group is called the *annihilator* of χ and is denoted by $\mathrm{Ann}(\chi)$.

Similarly suppose H is a subgroup of G. Then if $\chi \in \widehat{G}$ is a character of G such that $\chi(h) = 1$ for all $h \in H$ then $\overline{\chi}(h) = 1$ for all $h \in H$. Hence the set of all characters $\chi \in \widehat{G}$ such that $\chi(h) = 1$ for all $h \in H$ forms a group. This group is called the *annihilator* of H and is denoted by $\mathrm{Ann}(H)$. Since in both cases the notation Ann is used usually the context will make it clear whether we are talking about the annihilator of a character or the annihilator of a subgroup.

Definition 3.1.3: Annihilator

Let G be a finite abelian group. For any character $\chi \in \widehat{G}$, the annihilator *of χ, $\mathrm{Ann}(\chi) = \{g \in G : \chi(g) = 1\}$ which is a subgroup of G.*

For any subgroup H of G, the annihilator *of H, $\mathrm{Ann}(H) = \{\chi \in \widehat{G} : \chi(h) = 1 \text{ for all } h \in H\}$ which is a subgroup of $\widehat{G}$.*

Now every element of $\widehat{G}$ is a function χ which takes an element of G and maps to a complex number in S^1. Similarly we can think of every element of as a function from $\widehat{G}$ to S^1. In fact for every element $g \in G$ define the homomorphism $\mathrm{ev}_g : \widehat{G} \to S^1$ where $\mathrm{ev}_g(\chi) = \chi(g)$. Hence ev_g is a character of $\widehat{G}$ for all $g \in G$.

Theorem 3.1.4

Let G be a finite abelian group.

(i) If $\chi \in \widehat{G}$ be any nontrivial character then

$$\sum_{g \in G} \chi(g) = 0 \tag{3.1}$$

(ii) If $g \in G$ with $g \neq 1_G$ then

$$\sum_{\chi \in \widehat{G}} \chi(g) = 0 \tag{3.2}$$

Proof: Since χ is nontrivial there exists $h \in G$ such that $\chi(h) \neq 1$. Then

$$\chi(h) \sum_{g \in G} \chi(g) = \sum_{g \in G} \chi(h \cdot g) = \sum_{g \in G} \chi(g)$$

Hence we have

$$(\chi(h) - 1) \sum_{g \in G} \chi(g) = 0$$

Since we know $\chi(h) \neq 1$ we have $\sum_{g \in G} \chi(g) = 0$.

Now from the discussion above for any $g \in G$ and $\chi \in hG$ we have $\chi(g) = \mathrm{ev}_g(\chi)$. Hence

$$\sum_{\chi \in \widehat{G}} \chi(g) = \sum_{\chi \in \widehat{G}} \mathrm{ev}_g(\chi)$$

Since $g \neq 1_G$, there is a $\chi \in \widehat{G}$ such that $\chi(g) \neq 1$. Hence ev_g is a nontrivial character of $\widehat{G}$. Hence by the first part of the theorem we have $\sum_{\chi \in \widehat{G}} \mathrm{ev}_g(\chi) = 0$. ■

The above theorem leaves the case when χ is the trivial character. Then for all $g \in G$, $\chi(g) = 1$. Hence we have $\sum_{g \in G} \chi(g) = |G|$. Hence we have the following corollary

Observation 3.4. *For any finite abelian group G. Then*

(i) Let χ is a character of G.

$$\sum_{g \in G} \chi(g) = \begin{cases} |G| & \text{if } \chi = \chi_0 \\ 0 & \text{otherwise} \end{cases} \tag{3.3}$$

(ii) $|G| = \sum_{\chi \in \widehat{G}} \sum_{g \in G} \chi(g)$

So now by counting in two ways for the sum $\sum_{\chi \in \widehat{G}} \sum_{g \in G} \chi(g)$ we get the following theorem:

Theorem 3.1.5

For any finite abelian group G, we have $|G| = |\widehat{G}|$.

Proof: From Observation 3.4(ii) we have

$$|G| = \sum_{\chi \in \widehat{G}} \sum_{g \in G} \chi(g) = \sum_{g \in G} \sum_{\chi \in \widehat{G}} \chi(g) = |\widehat{G}|$$

Hence we have the result. ■

From above theorem and Observation 3.4(i) we get the *orthogonality relations for characters.*

Observation 3.5. *Let G be a finite abelian group.*

(i) Let χ, τ are characters of G. Then

$$\frac{1}{|G|} \sum_{g \in G} \chi(g) \cdot \overline{\tau(g)} = \begin{cases} 1 & \text{if } \chi = \tau \\ 0 & \text{otherwise} \end{cases}$$

(ii) If $g, h \in G$ are any two elements then

$$\frac{1}{|G|} \sum_{\chi \in \widehat{G}} \chi(g) \cdot \overline{\chi(h)} = \begin{cases} 1 & \text{if } g = h \\ 0 & \text{otherwise} \end{cases}$$

Since we already showed that for every element $g \in G$, ev_g is a character of $\widehat{G}$. Theorem 3.1.5 gives us the possibility that G and $\widehat{G}$ are isomorphic. In Lemma 3.1.1 we already showed that for cyclic groups G and $\widehat{G}$ are isomorphic. In the next few results we will show that this is the case for all finite abelian groups.

By Fundamental Theory of Finite Abelian Groups every finite abelian group G can be written as a direct product of finitely many cyclic groups. Hence it is enough to show that if G_1 and G_2 are two finite abelian groups and $G = G_1 \times G_2$ then $\widehat{G}$ is isomorphic to the direct product of $\widehat{G}_1$ and $\widehat{G}_2$.

Theorem 3.1.6

Let G_1 and G_2 be two finite abelian groups and let $G = G_1 \times G_2$. Then $\widehat{G} \cong \widehat{G}_1 \times \widehat{G}_2$.

Proof: Since G is the direct product of G_1 and G_2, G consists of pairs (g_1, g_2) where $g_1 \in G_1$ and $g_2 \in G_2$. So consider the map $\varphi : \widehat{G}_1 \times \widehat{G}_2 \to \widehat{G}$ where for any $\chi_1 \in \widehat{G}_1$ and $\chi_2 \in \widehat{G}_2$, $\chi := \varphi(\chi_1, \chi_2)$ is the character of G defined by the following:

$$\text{for all } (g_1, g_2) \in G \qquad \chi(g_1, g_2) = \chi(g_1, 1) \cdot \chi(1, g_2) = \chi_1(g_1) \cdot \chi_2(g_2)$$

Hence φ is a homomorphism. Now we will show that φ is an isomorphism.

Let $\chi \in \widehat{G}$. Then define $\chi_1(g_1) := \chi(g_1, 1)$ for all $g_1 \in G_1$ and similarly define $\chi_2(g_2) := \chi(1, g_2)$ for all $g_2 \in G_2$. Then χ_1 is a character of G_1 and χ_2 is a character of G_2. And also for any $(g_1, g_2) \in G$ we have

$$\chi_1(g_1) \cdot \chi_2(g_2) = \chi(g_1, 1) \cdot \chi(1, g_2) = \chi(g_1, g_2)$$

Hence $\varphi(\chi_1, \chi_2) = \chi$. Hence φ is an isomorphism. Therefore $\widehat{G} \cong \widehat{G}_1 \times \widehat{G}_2$. ■

Theorem 3.1.7

For any finite abelian group G, G is isomorphic to $\widehat{G}$.

Proof: By Fundamental Theorem of Finite Abelian Groups, G can be written as a direct product of finitely many cyclic groups i.e. there exists cyclic groups $Z_{n_1}, Z_{n_2}, \ldots, Z_{n_k}$ such that

$$G \cong Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_k}$$

Hence by repeated use of Theorem 3.1.6 we have $\widehat{G} \cong \widehat{Z}_{n_1} \times \widehat{Z}_{n_2} \times \cdots \times \widehat{Z}_{n_k}$. Since by Lemma 3.1.1 for all $i \in \{1, 2, \ldots, k\}$, $\widehat{Z}_{n_i} \cong Z_{n_i}$ we have $\widehat{G} \cong Z_{n_1} \times Z_{n_2} \times \cdots \times Z_{n_k} \cong G$ Hence we have the result. ■

3.2 Characters of Finite Fields

In a finite field $\mathbb{F}_q$ there are two inherent group structures–namely, the additive group and the multiplicatve group. The nonzero elements of $\mathbb{F}_q$ forms a cyclic group of $q-1$ elements under multiplication and the whole $\mathbb{F}_q$ forms a group of q elements under addition.

3.2.1 Additive Characters of $\mathbb{F}_q$

We first turn to the additive group of $\mathbb{F}_q$. Characters of the additive group $\mathbb{F}_q$ are called *additive characters* of $\mathbb{F}_q$. We denote the set of all additive characters of $\mathbb{F}_q$ by $\boldsymbol{\chi}_q$.

Suppose p be the characteristic of $\mathbb{F}_q$. Then the prime field contained in $\mathbb{F}_q$ is $\mathbb{F}_p$. Let $\mathrm{Tr} : \mathbb{F}_q \to \mathbb{F}_p$ be the absolute trace function from $\mathbb{F}_q$ to $\mathbb{F}_p$. Then define the function χ_1 by

$$\chi_1(\alpha) = e^{2\pi i \frac{\mathrm{Tr}(\alpha)}{p}} \quad \forall\, \alpha \in \mathbb{F}_q$$

Then χ_1 is an additive character of $\mathbb{F}_q$ since for all $\alpha_1, \alpha_2 \in \mathbb{F}_q$, $\mathrm{Tr}(\alpha_1 + \alpha_2) = \mathrm{Tr}(\alpha_1) + \mathrm{Tr}(\alpha_2)$ and hence $\chi_1(\alpha_1 + \alpha_2) = \chi_1(\alpha_1) \cdot \chi_1(\alpha_2)$. This character χ_1 is called the *canonical additive character* of $\mathbb{F}_q$.

Theorem 3.2.1

For $\beta \in \mathbb{F}_q$, define the function χ_β with $\chi_\beta(\alpha) = \chi_1(\beta \cdot \alpha)$ for all $\alpha \in \mathbb{F}_q$. Then χ_β is an additive character of $\mathbb{F}_q$ and every additive character of $\mathbb{F}_q$ is obtained this way i.e. for all $\chi \in \boldsymbol{\chi}_q$, there exists $\beta \in \mathbb{F}_q$ such that for all $\alpha \in \mathbb{F}_q$ we

have $\chi(\alpha) = \chi_\beta(\alpha)$.

Proof: Now for any $\alpha_1, \alpha_2 \in \mathbb{F}_q$, we have

$$\chi_\beta(\alpha_1 + \alpha_2) = \chi_1(\beta(\alpha_1 + \alpha_2)) = \chi_1(\beta \cdot \alpha_1) \cdot \chi_1(\beta \cdot \alpha_2) = \chi_\beta(\alpha_1) \cdot \chi_\beta(\alpha_2)$$

Hence χ_β is an additive character of $\mathbb{F}_q$.

Now we will show that for any two distinct $\beta_1, \beta_2 \in \mathbb{F}_q$ with $\beta_1 \neq \beta_2$ there exists an $\alpha \in \mathbb{F}_q$ such that $\chi_{\beta_1}(\alpha) \neq \chi_{\beta_2}(\alpha)$. Suppose that is not the case. Then for all $\alpha \in \mathbb{F}_q$,

$$\chi_{\beta_2}(\alpha) = \chi_{\beta_2}(\alpha) \iff \chi_1(\beta_1 \cdot \alpha) = \chi_1(\beta_2 \cdot \alpha) \iff \chi_1((\beta_1 - \beta_2)\alpha) = 1.$$

Now $\chi_1((\beta_1 - \beta_2)\alpha) = 1$ for all $\alpha \in \mathbb{F}_q$ if and only if $\mathrm{Tr}((\beta_1 - \beta_2)\alpha) = 0$ for all $\alpha \in \mathbb{F}_q$. Since $\mathrm{Tr}(X) = X + X^p + \cdots + X^{p^{r-1}}$ where $q = p^r$. Since Tr is $\mathbb{F}_p$–linear, $\mathrm{Tr}((\beta_1 - \beta_2)X)$ has p^r roots in $\mathbb{F}_q$ which is more than the degree p^{r-1}. Hence $\mathrm{Tr}((\beta_1 - \beta_2)X)$ is the zero polynomial. Therefore $\beta_1 = \beta_2$. So for any two distinct $\beta_1, \beta_2 \in \mathbb{F}_q$ with $\beta_1 \neq \beta_2$ there exists an $\alpha \in \mathbb{F}_q$ such that $\chi_{\beta_1}(\alpha) \neq \chi_{\beta_2}(\alpha)$.

Hence for every $\beta \in \mathbb{F}_q$ we have a distinct additive character. Therefore we have q additive characters. Since there are q additive characters of $\mathbb{F}_q$ by Theorem 3.1.5 every additive character of $\mathbb{F}_q$ can be obtained the above way. ■

The orthogonality relations in Observation 3.5 when applied to additive characters and along with Theorem 3.2.1 gives the following result.

Theorem 3.2.2

(i) For any $a \in \mathbb{F}_q$, *the corresponding additive character is* χ_a. *Then*

$$\sum_{\gamma \in \mathbb{F}_q} \chi_a(\gamma) = \begin{cases} 0 & \text{if } a \neq 0 \\ q & \text{if } a = 0 \end{cases}$$

(ii) For any two additive characters χ_a, χ_b *where* $a, b \in \mathbb{F}_q$,

$$\sum_{\gamma \in \mathbb{F}_q} \chi_a(\gamma) \cdot \overline{\chi_b}(\gamma) = \begin{cases} 0 & \text{if } a \neq b \\ q & \text{if } a = b \end{cases}$$

(iii) For any two elements $\alpha, \beta \in \mathbb{F}_q$,

$$\sum_{c \in \mathbb{F}_q} \chi_c(\alpha) \cdot \overline{\chi_c(\beta)} = \begin{cases} 0 & \text{if } \alpha \neq \beta \\ q & \text{if } \alpha = \beta \end{cases}$$

3.2.2 Multiplicative Characters of $\mathbb{F}_q$

Characters of the multiplicate group $\mathbb{F}_q^*$ are called *multiplicative characters* of $\mathbb{F}_q$. We denote the set of all multiplicative characters of $\mathbb{F}_q$ by $\boldsymbol{M}_q$. Since $\mathbb{F}_q^*$ is a cyclic group of order $q-1$ by Lemma 3.1.1 we have the following:

Lemma 3.2.3

The group of multiplicative characters of $\mathbb{F}_q$ *is a cycic group of order* $q-1$ *and are given by* $\{\psi_j : 0 \leq j \leq q-2\}$ *where if* g *is the generator of* $\mathbb{F}_q^*$ *then for all* $k \in \{0, 1, \ldots, q-2\}$ *we have* $\psi_j(g^k) = e^{2\pi i \frac{jk}{q-1}}$ *where* ψ_0 *is the identity element*

of the group and the trivial character.

Every multiplicative character ψ of $\mathbb{F}_q$ has the property that $\psi^{q-1} = \psi_0$. Let ψ is generator of the cyclic group of multiplicative characters. Then ψ is called the *principle character*. For any multiplicative character ψ of $\mathbb{F}_q$ we say ψ is of *order* d if $\psi^d = \psi_0$ and d is the smallest positive integer with this property. We use $\operatorname{ord}(\psi)$ to denote the order of ψ. Hence $d \mid q-1$. We say ψ is of *exponent* e is $\psi^e = \psi_0$. Clearly $d \mid e$ where d is the order of ψ. Let $\mathcal{M}_q^{(e)}$ denote the set of all multiplicative characters of $\mathcal{M}_q$ of exponent e.

Remark: For odd q a special multiplicative character of interest is the quadratic character. It is denoted by η. It only takes the value ± 1. So $\operatorname{ord}\eta = 2$. For all $\alpha \in \mathbb{F}_q^*$, $\eta(\alpha) = 1$ if α is a quadratic residue and -1 if α is a non-quadratic residue. Therefore for all $\alpha \in \mathbb{F}_q$, we have $\eta(\alpha) = \alpha^{(q-1)/2}$. The *Legendre* symbol $\left(\frac{\alpha}{q}\right)$ for any $\alpha \in \mathbb{F}_q$ is defined by $\eta(\alpha) = \left(\frac{\alpha}{q}\right)$

For any multiplicative character ψ of $\mathbb{F}_q$, the value of $\psi(-1)$ is of special interest. We obviously have $\psi(-1) = \pm 1$. Let $\operatorname{ord}(\psi) = m$. Then $m \mid q-1$. Hence values of ψ are m^{th} roots of unity. Therefore $\psi(-1)$ can be -1 if m is even. Let g be a primitive element of $\mathbb{F}_q$ i.e. $\mathbb{F}_q^*$ is generated by g. Hence $\psi(g) = \zeta_m$ where ζ_m is the mth root of unity. If m is even

$$\psi(-1) = \psi\left(g^{(q-1)/2}\right) = \zeta_m^{(q-1)/2}$$

Now

$$\zeta_m^{(q-1)/2} = -1 \iff \frac{q-1}{2} \equiv \frac{m}{2} \bmod m \iff \frac{q-1}{m} \equiv 1 \bmod 2$$

So m should be even and $\frac{q-1}{m}$ should be odd. Therefore we have the following two observations.

Observation 3.6. *For any multiplicative character $\psi \in \mathcal{M}_q$, $\psi(-1) = -1$ if and only if $\frac{q-1}{m}$ is odd and m is even.*

Observation 3.7. *If $\psi \in \mathcal{M}_q$ is the principle multiplicative character of $\mathbb{F}_q$ then $\psi(-1) = -1$.*

Suppose $d \mid q-1$. Then for every multiplicative character ψ of exponent d and for every $\alpha \in \mathbb{F}_q^*$, we have $\psi(\alpha^d) = \psi^d(\alpha) = 1$. Hence $\psi(\alpha) = 1$ if $\alpha \in (\mathbb{F}_q^*)^{(d)}$. Conversely if $\psi(\alpha) = 1$ for every $\alpha \in (\mathbb{F}_q^*)^{(d)}$ then $\psi^d = \psi_0$. Hence ψ is a character of exponent d then $\psi(\alpha)$ depends only on the cosets of $\mathbb{F}_q^* / (\mathbb{F}_q^*)^{(d)}$. Therefore there are precisely d such characters of exponent d.

Theorem 3.2.4

Suppose $d \mid q-1$. Then

$$\sum_{\psi \in \mathcal{M}_q^{(d)}} \chi(\alpha) = \begin{cases} d & \text{if } \alpha \in (\mathbb{F}_q^*)^{(d)} \\ 0 & \text{otherwise} \end{cases}$$

Proof: The characters of exponent d are characters of the group $\mathbb{F}_q^* / (\mathbb{F}_q^*)^{(d)}$. Hence this follows from Observation 3.4. ■

Again, the orthogonality relations in Observation 3.5 when applied to multiplicative characters gives the following result.

Theorem 3.2.5

(i) For any multiplicative character ψ of $\mathbb{F}_q$

$$\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha) = \begin{cases} 0 & \text{if } \psi \neq \psi_0 \\ q-1 & \text{if } \psi = \psi_0 \end{cases}$$

(ii) For any two multiplicative characters ψ and λ of $\mathbb{F}_q$,

$$\sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha) \cdot \overline{\lambda(\alpha)} = \begin{cases} 0 & \text{if } \psi \neq \lambda \\ q-1 & \text{if } \psi = \lambda \end{cases}$$

(iii) For any two elements $\alpha, \beta \in \mathbb{F}_q^*$,

$$\sum_{\psi \in \mathcal{M}_q} \psi(\alpha) \cdot \overline{\psi(\beta)} = \begin{cases} 0 & \text{if } \alpha \neq \beta \\ q-1 & \text{if } \alpha = \beta \end{cases}$$

Therefore the multiplicative characters form an orthonormal basis on all complex valued functions from $\mathbb{F}_q^*$ to $\mathbb{C}$. So we have the observation

Observation 3.8. *obs:mult-char-fourier-basis The set of multiplicative characters $\mathcal{M}_q$ of $\mathbb{F}_q$ forms an orthonormal basis for the vector space W of all complex valued functions $\varphi\colon \mathbb{F}_q^* \to \mathbb{C}$.*

Usually we extend the definition of multiplicative characters to the whole $\mathbb{F}_q$ so that we can define special character sums over the whole field. Let ψ is a multiplicative character then $\psi(0)$ is defined to be 0 if ψ is nontrivial multiplicative character and $\psi(0) = 1$ if ψ is the trivial character.

3.2.3 Lifting of Characters to Finite Extension

Let E be a finite extension field of $\mathbb{F}_q$. Let χ_1 and μ_1 be the canonical additive characters of $\mathbb{F}_q$ and E respectively. Let $\mathrm{Tr}_{E/\mathbb{F}_q}$ be the trace function from E to $\mathbb{F}_q$.

Then χ_1 and μ_1 are related by the identity

$$\chi_1(Tr_{E/\mathbb{F}_q}(\alpha)) = \mu_1(\alpha) \quad \forall\, \alpha \in E$$

Since the trace function satisfies transitivity property we have $\mathrm{Tr}_E = \mathrm{Tr}_{\mathbb{F}_q} \circ \mathrm{Tr}_{E/\mathbb{F}_q}$. Hence for all $\alpha \in E$, $\mu_1(\alpha) = \chi_1(\mathrm{Tr}_{E/\mathbb{F}_q}(\alpha))$.

Similarly if ψ_1 and λ_1 are principle multiplicative characters of $\mathbb{F}_q$ and E respectively then,

$$\lambda_1(\alpha) = \psi_1(\mathrm{N}_{E/\mathbb{F}_q}(\alpha)) \quad \forall\, \alpha \in E^*$$

where $\mathrm{N}_{E/\mathbb{F}_q}$ is the norm function from E to $\mathbb{F}_q$.

In the opposite way for any additive character χ of $\mathbb{F}_q$ or multiplicative character ψ of $\mathbb{F}_q$ we can lift the characters to an additive and multiplicative character of E respectively by taking $\chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$ and $\psi \circ \mathrm{N}_{E/\mathbb{F}_q}$ respectively. We denote these by $\chi^{(r)} := \chi \circ \mathrm{Tr}_{E/\mathbb{F}_q}$ and $\psi^{(r)} := \psi \circ \mathrm{N}_{E/\mathbb{F}_q}$ where $r = [E : \mathbb{F}_q]$.

In the following sections we will be working with hybrid character sums which are formed by taking product of multiple characters of both kind. Hence we will be using χ, τ to denote additive characters and ψ, λ to denote multiplicative characters.

3.3 Gaussian Sums

Having studied additive and multiplicative characters of $\mathbb{F}_q$ separately, we now turn to hybrid sums that interleave both types. The first and most fundamental such object is the *Gaussian sum.* A Gaussian sum is formed by taking the product of a multiplicative character and an additive character and summing over all nonzero elements of $\mathbb{F}_q$. The interplay between the two kinds of characters gives remarkable algebraic and analytic properties, and they will serve as the key tool throughout this chapter. We define them as follows.

Definition 3.3.1: Gaussian Sum

Let ψ be a multiplicative character and χ be an additive character of $\mathbb{F}_q$. Then the Gaussian Sum *associated to ψ and χ is defined as*

$$G(\psi, \chi) = \sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha) \cdot \chi(\alpha)$$

Hence the absolute value of the Gaussian sum is at most $q - 1$. But later we will see that the Gaussian sum in most cases is much smaller, $\sqrt{q}$ to be precise.

Theorem 3.3.1

Let ψ be a multiplicative character and χ be an additive character of $\mathbb{F}_q$. Then

$$G(\psi, \chi) = \begin{cases} q - 1 & \text{if } \psi = \psi_0, \chi = \chi_0 \\ -1 & \text{if } \psi = \psi_0, \chi \neq \chi_0 \\ 0 & \text{if } \psi \neq \psi_0, \chi = \chi_0 \end{cases} \tag{3.4}$$

If ψ is non-trivial multiplicative character and χ is non-trivial additive character then $|G(\psi, \chi)| = \sqrt{q}$.

Proof: If $\psi = \psi_0$ and $\chi = \chi_0$ then for all $\alpha \in \mathbb{F}_q^*$ we have $\psi(\alpha) = 1$ and $\chi(\alpha) = 1$. Hence $G(\psi_0, \chi_0) = q - 1$.

Now suppose $\psi = \psi_0$ and χ is non-trivial additive character. Then by Theorem 3.2.2(i)

$$G(\psi_0, \chi) = \sum_{\alpha \in \mathbb{F}_q^*} \chi(\alpha) = \sum_{\alpha \in \mathbb{F}_q} \chi(\alpha) - \chi(0) = -1$$

Let ψ be a non-trivial multiplicative character and $\chi = \chi_0$. Then for all $\alpha \in \mathbb{F}_q^*$ we have $\chi(\alpha) = 1$. Hence by Theorem 3.2.5(i)

$$G(\psi, \chi_0) = \sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha) = 0$$

So now assume both ψ and χ are non-trivial multiplicative and additive characters of $\mathbb{F}_q$ respectively. Then we

have

$$\begin{aligned}
|G(\psi,\chi)|^2 &= \overline{G(\psi,\chi)}\cdot G(\psi,\chi)\\
&= \left(\sum_{\alpha\in\mathbb{F}_q^*}\overline{\psi(\alpha)}\cdot\overline{\chi(\alpha)}\right)\cdot\left(\sum_{\beta\in\mathbb{F}_q^*}\psi(\beta)\cdot\chi(\beta)\right)\\
&= \sum_{\alpha\in\mathbb{F}_q^*}\sum_{\beta\in\mathbb{F}_q^*}\overline{\psi(\alpha)}\cdot\overline{\chi(\alpha)}\cdot\psi(\beta)\cdot\chi(\beta)\\
&= \sum_{\alpha\in\mathbb{F}_q^*}\sum_{\beta\in\mathbb{F}_q^*}\psi(\beta\cdot\alpha^{-1})\cdot\chi(\beta-\alpha)\\
&= \sum_{\alpha\in\mathbb{F}_q^*}\sum_{\gamma\in\mathbb{F}_q^*}\psi(\gamma)\cdot\chi(\alpha(\gamma-1)) && [\text{Take } \gamma=\beta\cdot\alpha^{-1}]\\
&= \sum_{\gamma\in\mathbb{F}_q^*}\psi(\gamma)\cdot\left(\sum_{\alpha\in\mathbb{F}_q^*}\chi(\alpha(\gamma-1))\right)\\
&= \sum_{\gamma\in\mathbb{F}_q^*}\psi(\gamma)\cdot\left(\sum_{\alpha\in\mathbb{F}_q}\chi(\alpha(\gamma-1))\right)-\chi(0)\sum_{\gamma\in\mathbb{F}_q^*}\psi(\gamma)\\
&= \sum_{\gamma\in\mathbb{F}_q^*}\psi(\gamma)\cdot\left(\sum_{\alpha\in\mathbb{F}_q}\chi(\alpha(\gamma-1))\right) && [\text{By } \textit{Theorem } 3.2.5(i)]
\end{aligned}$$

Hence by Theorem 3.2.2(i)

$$\sum_{\alpha\in\mathbb{F}_q}\chi(\alpha(\gamma-1)) = \begin{cases}0 & \text{if } \gamma\neq 1\\ q & \text{if } \gamma = 1\end{cases}$$

Therefore $|G(\psi,\chi)|^2 = \psi(1)q = q$ and so we have the required result. ■

Now under various transformations of the additive or multiplicative characters Gaussian sums gives some useful identities. We will see some of these below.

Lemma 3.3.2

Gaussian sums for the finite field $\mathbb{F}_q$ satisfy the following identities. For any $a\in\mathbb{F}_q$ let χ_a be the additive character defined as in Theorem 3.2.1. Let $\psi\in\boldsymbol{\mathcal{M}}_q$ be any multiplicative character and $\chi\in\boldsymbol{\mathcal{X}}_q$ be any additive character of $\mathbb{F}_q$. Then we have the following identities

(i) $G(\psi,\chi_{ab}) = \overline{\psi(a)}\cdot G(\psi,\chi_b)$ *for any* $a\in\mathbb{F}_q^*$ *and* $b\in\mathbb{F}_q$.

(ii) $G(\psi,\overline{\chi}) = \psi(-1)\cdot G(\psi,\chi)$.

(iii) $G(\overline{\psi},\chi) = \psi(-1)\overline{G(\psi,\chi)}$.

(iv) $G(\psi,\chi)\cdot G(\overline{\psi},\chi) = \psi(-1)q$ *if* ψ *and* χ *are non-trivial multiplicative and additive characters of* $\mathbb{F}_q$ *respectively.*

(v) $G(\psi^p,\chi_b) = G(\psi,\chi_{\sigma(b)})$ *for any* $b\in\mathbb{F}_q$ *where* $p=\mathrm{char}(\mathbb{F}_q)$ *and* $\sigma(X)=X^p$ *be the Frobenius map of* $\mathbb{F}_q$.

Proof:

(i) For any $\gamma\in\mathbb{F}_q$ we have $\chi_{ab}(\gamma)=\chi_1(ab\cdot\gamma)=\chi_b(a\cdot\gamma)$. Hence

$$G(\psi,\chi_{ab}) = \sum_{\gamma\in\mathbb{F}_q^*}\psi(\gamma)\cdot\chi_{ab}(\gamma) = \sum_{\gamma\in\mathbb{F}_q^*}\psi(\gamma)\cdot\chi_b(a\cdot\gamma)$$

Now take $d = a \cdot \gamma$. Then $\gamma = a^{-1} \cdot d$. Hence

$$G(\psi, \chi_{ab}) = \sum_{d \in \mathbb{F}_q^*} \psi(a^{-1} \cdot d) \cdot \chi_b(d) = \psi(a^{-1}) \sum_{d \in \mathbb{F}_q^*} \psi(d) \cdot \chi_b(d) = \overline{\psi(a)} \cdot G(\psi, \chi_b)$$

Hence we have the first identity.

(ii) Now we have $\chi = \chi_b$ for some suitable $b \in \mathbb{F}_q$. Therefore $\overline{\chi} = \chi_{(-1)\cdot b}$. So by the first identity we have $G(\psi, \overline{\chi}) = \overline{\psi(-1)} \cdot G(\psi, \chi_b) = \psi(-1) \cdot G(\psi, \chi)$. The last equality holds because $\psi(-1) = \pm 1$.

(iii) Now again we have $\overline{\chi} = \chi_b$ for some suitable $b \in \mathbb{F}_q$. Hence $\chi = \chi_{(-1)\cdot b}$. Again by using the first identity we have

$$G(\overline{\psi}, \chi) = G\left(\overline{\psi}, \chi_{(-1)\cdot b}\right) = \overline{\psi(-1)} \cdot G(\overline{\psi}, \chi_b) = \psi(-1) \cdot G(\overline{\psi}, \overline{\chi})$$

Now from the definition of Gaussian sums we have $G(\overline{\psi}, \overline{\chi}) = \overline{G(\psi, \chi)}$. Hence we have the required identity.

(iv) Using part (ii) and (iii) we have

$$G(\psi, \chi) \cdot G(\overline{\psi}, \chi) = G(\psi, \chi) \cdot \psi(-1)\overline{G(\psi, \chi)} = \psi(-1) \cdot |G(\psi, \chi)|^2$$

Since ψ and χ are non-trivial multiplicative and additive characters of $\mathbb{F}_q$ respectively by Theorem 3.3.1 we have $|G(\psi, \chi)|^2 = q$. Hence we have the required identity.

(v) If Tr is the absolute trace function from $\mathbb{F}_q$ to $\mathbb{F}_p$ then for any $\alpha \in \mathbb{F}_q$ we have $\mathrm{Tr}(\alpha) = \mathrm{Tr}(\alpha^p)$. Hence $\chi_1(\alpha) = \chi_1(\alpha^p)$. Hence for any $\alpha \in \mathbb{F}_q$, we get $\chi_b(\alpha) = \chi_1(b \cdot \alpha) = \chi_1(b^p \cdot \alpha^p) = \chi_{\sigma(b)}(\alpha^p)$. Therefore

$$G(\psi^p, \chi_b) = \sum_{\alpha \in \mathbb{F}_q^*} \psi^p(\alpha) \cdot \chi_b(\alpha) = \sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha^p)\chi_{\sigma(b)}(\alpha^p) = G(\psi, \chi_{\sigma(b)})$$

Therefore we have the required result.

■

Now we can use Gaussian Sums in a variety of context. For any multiplicative character ψ we can express the value of ψ at any nonzero $\alpha \in \mathbb{F}_q$ using Gaussian sums and additive characters. And similarly for any additive character χ of $\mathbb{F}_q$ we can express the value of χ at any nonzero $\alpha \in \mathbb{F}_q$ using Gaussian sums and multiplicative characters.

Now by Theorem 3.2.2(iii) for any $\alpha \in \mathbb{F}_q^*$ we have

$$\begin{aligned}
\psi(\alpha) &= \frac{1}{q} \sum_{\beta \in \mathbb{F}_q^*} \psi(\beta) \sum_{c \in \mathbb{F}_q} \chi_c(\alpha)\overline{\chi_c(\beta)} \\
&= \frac{1}{q} \sum_{c \in \mathbb{F}_q} \chi_c(\alpha) \sum_{\beta \in \mathbb{F}_q^*} \psi(\beta) \cdot \overline{\chi}_c(\beta) \\
&= \frac{1}{q} \sum_{c \in \mathbb{F}_q} \chi_c(\alpha) \cdot G(\psi, \overline{\chi}_c) \\
&= \frac{1}{q} \sum_{\chi \in \boldsymbol{\chi}_q} G(\psi, \overline{\chi})\chi(\alpha)
\end{aligned}$$

This may be thought of as the Fourier expansion of ψ in terms of the additive characters of $\mathbb{F}_q$, with Gaussian sums appearing as Fourier coefficients.

We can do similar thing for multiplicative characters of $\mathbb{F}_q$. By using Theorem 3.2.5(iii) we get

$$\chi(\alpha) = \frac{1}{q-1} \sum_{\psi \in \boldsymbol{M}_q} G(\overline{\psi}, \chi)\chi(\alpha)$$

This can be interpreted as the Fourier expansion of the multiplicative characters of $\mathbb{F}_q$ with the Gaussian sums as Fourier coefficients. Hence we have the observations:

Observation 3.9. *For any multiplicative character ψ and additive character χ of $\mathbb{F}_q$ and any $\alpha \in \mathbb{F}_q^*$ we have*

(i) $\psi(\alpha) = \frac{1}{q} \sum_{\chi \in \boldsymbol{\chi}_q} G(\psi, \overline{\chi})\chi(\alpha)$

(ii) $\chi(\alpha) = \frac{1}{q-1} \sum_{\psi \in \boldsymbol{M}_q} G(\overline{\psi}, \chi)\psi(\alpha)$

We now a another special formula for Gaussian sums for 2-degree extension of finite fields and it has a lot of application. But this has a restriction on the underlying field.

Theorem 3.3.3 Stickelberger's Theorem

Let q be a prime power and $\psi \in \boldsymbol{M}_q$ be a non-trivial multiplicative character of $\mathbb{F}_{q^2}$ of order m such that $m \mid q-1$. Let χ_1 be the canonical additive character of $\mathbb{F}_{q^2}$. Then

$$G(\psi, \chi_1) = \begin{cases} q & \text{if } m \text{ is odd or } \frac{q+1}{m} \text{ is even} \\ -q & \text{if } m \text{ is even and } \frac{q+1}{m} \text{ is odd} \end{cases}$$

Proof: For brevity of notations we write $E = \mathbb{F}_{q^2}$. Let γ be a primitive element of E. So set $g = \gamma^{q+1}$. Therefore $g^{q-1} = 1$ and so we get g as primitive element of $\mathbb{F}_q$. Therefore for all $\alpha \in E^*$ there exists unique $j \in \{0, 1, \ldots, q-1$ and $k \in \{0, \ldots, q+1\}$ such that $\alpha = g^j \cdot \gamma^k$. Hence $\psi(g) = \psi^{q+1}(\gamma) = 1$ as $\operatorname{ord}(\psi) = m \mid q+1$. Therefore we have

$$\begin{aligned} G(\psi, \chi_1) &= \sum_{j=0}^{q-2} \sum_{k=0}^{q} \psi(g^j \cdot \gamma^k) \cdot \chi_1(g^j \cdot \gamma^k) \\ &= \sum_{k=0}^{q} \psi^k(\gamma) \sum_{j=0}^{q-2} \chi_1(g^j \cdot \chi^k) \\ &= \sum_{k=0}^{q} \psi^k(\gamma) \sum_{\alpha \in \mathbb{F}_q^*} \chi_1(\alpha \cdot \chi^k) \end{aligned}$$

Let τ_1 is the canonical additive character of $\mathbb{F}_q$ then for any $\alpha \in \mathbb{F}_q^*$ and $k \in \{0, 1, \ldots, q\}$ we get

$$\chi_1(\alpha \cdot \gamma^k) = \tau_1\left(\operatorname{Tr}_{E/\mathbb{F}_q}(\alpha \cdot \gamma^k)\right) = \tau_1\left(\alpha \cdot \operatorname{Tr}_{E/\mathbb{F}_q}(\gamma^k)\right)$$

Hence we get

$$\sum_{\alpha \in \mathbb{F}_q^*} \chi_1(\alpha \cdot \chi^k) = \sum_{\alpha \in \mathbb{F}_q^*} \tau_1\left(\alpha \cdot \operatorname{Tr}_{E/\mathbb{F}_q}(\gamma^k)\right) = \begin{cases} -1 & \text{if } \operatorname{Tr}_{E/\mathbb{F}_q}(\gamma^k) \neq 0 \\ q-1 & \text{if } \operatorname{Tr}_{E/\mathbb{F}_q}(\gamma^k) = 0 \end{cases}$$

Now since $[E : \mathbb{F}_q] = 2$ we have $\operatorname{Tr}_{E/\mathbb{F}_q}(\gamma^k) = \gamma^k + \gamma^{k \cdot q}$. Therefore

$$\operatorname{Tr}_{E/\mathbb{F}_q}(\gamma^k) = 0 \iff \gamma^k = -\gamma^{k \cdot q} \iff \gamma^{k(q-1)} = -1$$

So if q is odd then $\gamma^{k(q-1)} = -1 \iff k = \frac{q+1}{2}$ and therefore we have

$$\sum_{\alpha \in \mathbb{F}_q^*} \chi_1(\alpha \cdot \chi^k) = \begin{cases} -1 & \text{if } k \neq \frac{q+1}{2} \\ q-1 & \text{if } k = \frac{q+1}{2} \end{cases}$$

And by using Theorem 3.2.2

$$G(\psi, \chi_1) = q \cdot \psi^{q+1/2}(\gamma) - \sum_{k=0}^{q} \psi^k(\gamma) = q \cdot \psi^{(q+1)/2}$$

Recall that ψ has order m and $m \mid q+1$. Therefore $\psi^{(q+1)/2}(\gamma) = \pm 1$. Now if $\frac{q+1}{m}$ is even then we can rewrite

$$\psi^{(q+1)/2}(\gamma) = (\psi^m(\gamma))^{(q+1)/2m} = 1$$

So suppose $\frac{q+1}{m}$ is odd, say $\frac{q+1}{m} = 2\ell + 1$ for some $\ell \geq 0$. Since q is odd, $q+1$ is even, and as $(q+1)/m$ is odd the factor of 2 in $q+1$ must lie entirely in m; in particular m is even. Writing $\frac{q+1}{2} = m\ell + \frac{m}{2}$, we get

$$\psi^{(q+1)/2}(\gamma) = (\psi^m(\gamma))^{\ell} \cdot \psi^{m/2}(\gamma) = \psi^{m/2}(\gamma).$$

Now $\psi^{m/2}(\gamma)$ is a square root of unity since order of $\psi = m$ and γ is primitive element of E we have $\psi^{(q+1)/2}(\gamma) = -1$. Therefore $G(\psi, \chi_1) = (-1)^{(q+1)/m} \cdot q$. So we have the theorem. ■

We can give a relation between Gaussian sums a multiplicative and additive character and Gaussian sum of lifted characters. To build that relation we need the theory of L-functions in section 3.5. Later we proved the Davenport-Hasse Theorem which gives this relation.

Before going into this in the next section we will define another hybrid sum called *Jacobi Sums*. Like Gaussian sum this also has many important applications. In the later sections we will use both Gaussian sums and Jacobi Sums to prove many interesting results

3.4 Jacobi Sums

Jacobi sums form another class of exponential sums which are of great importance in algebraic number theory. Jacobi sums are defined on product of many multiplicative characters.

Let $\psi \in \boldsymbol{M}_q$ is a multiplicative character of $\mathbb{F}_q$. So we extend the definition of ψ following subsection 3.2.2 by setting $\psi(0) = 1$ if ψ is the trivial character and $\psi(0) = 0$ if ψ is a non-trivial character. From now on by default we will always use this extended definition of multiplicative characters.

So now let $\lambda_1, \lambda_2, \ldots, \lambda_k \in \boldsymbol{M}_q$ be k multiplicative characters of $\mathbb{F}_q$. Let $\alpha \in \mathbb{F}_q$ be fixed. Then define the sum

$$J_\alpha(\lambda_1, \ldots, \lambda_k) = \sum_{\substack{\alpha_1, \ldots, \alpha_k \in \mathbb{F}_q \\ \alpha_1 + \cdots + \alpha_k = \alpha}} \lambda_1(\alpha_1) \cdot \lambda_2(\alpha_2) \cdots \lambda_k(\alpha_k)$$

Thus in the above sum there are q^{k-1} terms. Now if $\alpha \neq 0$ then define $\beta_i = \alpha^{-1} \cdot \alpha_i$. Then $\beta_1 + \cdots + \beta_k = 1$ and the above summations gives us

$$\begin{aligned} J_\alpha(\lambda_1, \ldots, \lambda_k) &= \sum_{\substack{\beta_1, \ldots, \beta_k \in \mathbb{F}_q \\ \beta_1 + \cdots + \beta_k = 1}} \lambda_1(\alpha \cdot \beta_1) \cdots \lambda_k(\alpha \cdot \beta_2) \\ &= (\lambda_1 \cdots \lambda_k)(\alpha) \sum_{\substack{\beta_1, \ldots, \beta_k \in \mathbb{F}_q \\ \beta_1 + \cdots + \beta_k = 1}} \lambda_1(\beta_1) \cdots \lambda_k(\beta_2) \\ &= (\lambda_1 \cdots \lambda_k)(\alpha) \cdot J_1(\lambda_1, \ldots, \lambda_k) \end{aligned}$$

Because of this relation it is enough to consider only the sums $J_0(\lambda_1, \ldots, \lambda_k)$ and $J_1(\lambda_1, \ldots, \lambda_k)$. The second sum is called the *Jacobi sum* and it has more important applications and so we use a slightly simpler notation.

Definition 3.4.1: Jacobi Sum

Let $\lambda_1, \ldots, \lambda_k \in \boldsymbol{M}_q$ be multiplicative characters of $\mathbb{F}_q$. Then the Jacobi sum is defined by

$$J(\lambda_1, \ldots, \lambda_k) \sum_{\substack{\alpha_1, \ldots, \alpha_k \in \mathbb{F}_q \\ \alpha_1 + \cdots + \alpha_k = 1}} \lambda_1(\alpha_1) \cdots \lambda_k(\alpha_k)$$

Now we want to find the value of $J(\lambda_1, \ldots, \lambda_k)$ and $J_0(\lambda_1, \ldots, \lambda_k)$. If $k = 1$ then $J(\lambda) = \lambda(1) = 1$ for any $\lambda \in \boldsymbol{\mathcal{M}}_q$ of $\mathbb{F}_q$. And $J_0(\lambda) = 0$ if λ is non-trivial and its 1 if λ is trivial multiplicative character. Therefore, Jacobi sums are only of interest when $k \geq 2$. Now first we have the following observation

Observation 3.10. *By the definition of the Jacobi sums $J(\lambda_1, \ldots, \lambda_k)$ and $J_0(\lambda_1, \ldots, \lambda_k)$ are independent of the order in which the characters λ_i are listed.*

Now we can evaluate Jacobi sums in some cases pretty easily. Specially if at least one of the multiplicative characters is trivial.

Lemma 3.4.1

Let $\lambda_1, \ldots, \lambda_k \in \boldsymbol{\mathcal{M}}_q$ are multiplicative characters of $\mathbb{F}_q$. Then

(i) If $\lambda_1 = \cdots = \lambda_k = \psi_0$ where ψ_0 is the trivial multiplicative character then

$$J(\lambda_1, \ldots, \lambda_k) = J_0(\lambda_1, \ldots, \lambda_k) = q^{k-1}$$

(ii) If some but not all λ_i's are trivial then

$$J(\lambda_1, \ldots, \lambda_k) = J_0(\lambda_1, \ldots, \lambda_k) = 0$$

Proof:

(i) Since all λ_i's are trivial for all $\alpha \in \mathbb{F}_q$, $\lambda(\alpha) = 1$. Hence each term of both the Jacobi sums give 1. Now the sum is over q^{k-1} choices of $(\alpha_1, \ldots, \alpha_k) \in \mathbb{F}_q^k$. Hence $J(\lambda_1, \ldots, \lambda_k) = J_0(\lambda_1, \ldots, \lambda_k) = q^{k-1}$.

(ii) WLOG suppose $\lambda_1, \ldots, \lambda_t$ are non-trivial multiplicative characters where $1 \leq t < k$ and $\lambda_{t+1}, \ldots, \lambda_k$ are all trivial. Then we have

$$\begin{aligned} J(\lambda_1, \ldots, \lambda_k) &= \sum_{\substack{\alpha_1, \ldots, \alpha_k \in \mathbb{F}_q \\ \alpha_1 + \cdots + \alpha_k = 1}} \lambda_1(\alpha_1) \cdots \lambda_k(\alpha_k) \\ &= \sum_{\substack{\alpha_1, \ldots, \alpha_k \in \mathbb{F}_q \\ \alpha_1 + \cdots + \alpha_k = 1}} \lambda_1(\alpha_1) \cdots \lambda_t(\alpha_t) \end{aligned}$$

Now for any fixed $\alpha_1, \ldots, \alpha_t \in \mathbb{F}_q$ there are q^{k-t-1} solutions of $(\alpha_{t+1}, \ldots, \alpha_k)$ such that $\alpha_{t+1} + \cdots + \alpha_k = 1 - (\alpha_1 + \cdots + \alpha_t)$. Therefore we get

$$\begin{aligned} J(\lambda_1, \ldots, \lambda_k) &= q^{k-h-1} \sum_{\alpha_1, \ldots, \alpha_t \in \mathbb{F}_q} \lambda_1(\alpha_1) \cdots \lambda_t(\alpha_t) \\ &= q^{k-h-1} \prod_{i=1}^{t} \left(\sum_{\alpha_i \in \mathbb{F}_q} \lambda_i(\alpha_i) \right) \\ &= 0 \qquad \text{[By } \textit{Theorem } 3.2.5(i)\text{]} \end{aligned}$$

With a very similar argument for $J_0(\lambda_1, \ldots, \lambda_k)$ we get the identity.

■

Before going to the case where all λ_i's are non-trivial we will show that a relation between $J_0(\lambda_1, \ldots, \lambda_k)$ and $J(\lambda_1, \ldots, \lambda_k)$. This will show that studying $J(\lambda_1, \ldots, \lambda_k)$ suffices to study all kinds of Jacobi sums.

Lemma 3.4.2

Let $\lambda_1, \ldots, \lambda_k \in \mathcal{M}_q$ are multiplicative characters of $\mathbb{F}_q$ with λ_k is non-trivial. Then

$$J_0(\lambda_1, \ldots, \lambda_k) = \begin{cases} 0 & \text{if } \prod_{i=1}^k \lambda_i \text{ is non-trivial} \\ \lambda_k(-1) \cdot (q-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1}) & \text{if } \prod_{i=1}^k \lambda_i \text{ is trivial} \end{cases}$$

Proof: For $k = 1$, we already showed $J_0(\lambda_1) = \lambda_1(0) = 0$ as λ_1 is non-trivial by hypothesis. So suppose $k \geq 2$. Then we have

$$\begin{aligned} J_0(\lambda_1, \ldots, \lambda_k) &= \sum_{\alpha \in \mathbb{F}_q} J_{-\alpha}(\lambda_1, \ldots, \lambda_{k-1}) \cdot \lambda_k(\alpha) \\ &= \sum_{\alpha \in \mathbb{F}_q} \left(\prod_{i=1}^{k-1} \lambda_i \right)(-\alpha) \cdot J(\lambda_1, \ldots, \lambda_{k-1}) \cdot \lambda_k(\alpha) \\ &= J(\lambda_1, \ldots, \lambda_{k-1}) \sum_{\alpha \in \mathbb{F}_q} \left(\prod_{i=1}^{k-1} \lambda_i \right)(-\alpha) \cdot \lambda_k(\alpha) \\ &= \left(\prod_{i=1}^{k-1} \lambda_i \right)(-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1}) \sum_{\alpha \in \mathbb{F}_q} \left(\prod_{i=1}^{k} \lambda_i \right)(\alpha) \end{aligned}$$

Now by Theorem 3.2.5(i) if $\prod_{i=1}^k \lambda_i$ is nontrivial then the sum 0, so the first case is shown and if $\prod_{i=1}^k \lambda_i$ is trivial then the last sum is $q-1$. Now since $\prod_{i=1}^k \lambda_i$ is trivial $\prod_{i=1}^{k-1} \lambda_i = \overline{\lambda_k}(-1) = \lambda_k(-1)$. So we have

$$J_0(\lambda_1, \ldots, \lambda_k) = \lambda_k(-1) \cdot (q-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1})$$

Hence we have the lemma. ■

Hence the only case for $J_0(\lambda_1, \ldots, \lambda_k)$ for which we still don't know the absolute value is the case when all λ_i's are non-trivial and $\prod_{i=1}^k \lambda_i$ is trivial. Since we understand Gaussian sums very well we will first build an important connection between Jacobi sums and Gaussian sums that will allow us to calculate the absolute value of Jacobi sums.

Theorem 3.4.3

Let $\lambda_1, \ldots, \lambda_k \in \mathcal{M}_q$ are non-trivial multiplicative characters of $\mathbb{F}_q$ and $\chi \in \mathcal{X}_q$ is a non-trivial additive character of $\mathbb{F}_q$.

(i) If $\prod_{i=1}^k$ is non-trivial then

$$J(\lambda_1, \ldots, \lambda_k) = \frac{G(\lambda_1, \chi) \cdot G(\lambda_2, \chi) \cdots G(\lambda_k, \chi)}{G(\lambda_1 \cdots \lambda_k, \chi)}$$

(ii) If $\prod_{i=1}^k$ is trivial then

$$\begin{aligned} J(\lambda_1, \ldots, \lambda_k) &= -\lambda_k(-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1}) \\ &= -\frac{1}{q} \prod_{i=1}^k G(\lambda_i, \chi) \end{aligned}$$

Proof: Since each λ_i's are non-trivial we have $\lambda_i(0) = 0$. By definition of

$$G(\lambda_i, \chi) = \sum_{\alpha_i \in \mathbb{F}_q^*} \lambda_i \alpha \cdot \chi(\alpha) = \sum_{\alpha_i \in \mathbb{F}_q} \lambda_i \alpha \cdot \chi(\alpha)$$

So now we will try to express the product of Gaussian sums as product of a Jacobi and a Gaussian sum.

$$\begin{aligned}
G(\lambda_1, \chi) \cdots G(\lambda_k, \chi) &= \prod_{i=1}^{k} \left(\sum_{\alpha_i \in \mathbb{F}_q} \lambda_i(\alpha_i) \cdot \chi(\alpha_i) \right) \\
&= \sum_{\alpha_1, \ldots, \alpha_k \in \mathbb{F}_q} \left(\prod_{i=1}^{k} \lambda_i(\alpha_i) \right) \cdot \chi \left(\sum_{i=1}^{k} \alpha_i \right) \\
&= \sum_{\alpha \in \mathbb{F}_q} \chi(\alpha) \sum_{\substack{\alpha_1, \ldots, \alpha_k \in \mathbb{F}_q \\ \alpha_1 + \cdots + \alpha_k = \alpha}} \left(\prod_{i=1}^{k} \lambda_i(\alpha_i) \right) \\
&= \sum_{\alpha \in \mathbb{F}_q} \chi(\alpha) \cdot J_\alpha(\lambda_1, \ldots, \lambda_k) \qquad (3.5)
\end{aligned}$$

Now if $\prod_{i=1}^{k} \lambda_i$ is non-trivial then by Lemma 3.4.2 we get $J_0(\lambda_1, \ldots, \lambda_k) = 0$. Now since for all $\alpha \in \mathbb{F}_q^*$ we have $J_\alpha(\lambda_1, \ldots, \lambda_k) = \left(\prod_{i=1}^{k} \lambda_i \right)(\alpha) \cdot J(\lambda_1, \ldots, \lambda_k)$. Therefore we have

$$\prod_{i=1}^{k} G(\lambda_i, \chi) = J(\lambda_1, \ldots, \lambda_k) \sum_{\alpha \in \mathbb{F}_q^*} \left(\prod_{i=1}^{k} \lambda_i \right)(\alpha) \cdot \chi(\alpha) = J(\lambda_1, \ldots, \lambda_k) \cdot G\left(\prod_{i=1}^{k} \lambda_i, \chi \right)$$

Since χ is non-trivial and $\prod_{i=1}^{k} \lambda_i$ is non-trivial we have $G(\lambda_1 \cdots \lambda_k, \chi) \neq 0$ by Theorem 3.3.1. So we have the first result.

Now if $\prod_{i=1}^{k} \lambda_i$ is trivial, then for all $\alpha \in \mathbb{F}_q^*$ we have $J_\alpha(\lambda_1, \ldots, \lambda_k) = J(\lambda_1, \ldots, \lambda_k)$. Hence rewriting the equation in (3.5) we get

$$\begin{aligned}
\prod_{i=1}^{k} G(\lambda_i, \chi) &= \sum_{\alpha \in \mathbb{F}_q} \chi(\alpha) \cdot J_\alpha(\lambda_1, \ldots, \lambda_k) \\
&= J_0(\lambda_1, \ldots, \lambda_k) + J(\lambda_1, \ldots, \lambda_k) \sum_{\alpha \in \mathbb{F}_q} \chi(\alpha) \\
&= J_0(\lambda_1, \ldots, \lambda_k) + (q-1) J(\lambda_1, \ldots, \lambda_k) && [\text{By } \textit{Theorem } 3.2.2(i)] \\
&= \sum_{\alpha \in \mathbb{F}_q} J(\lambda_1, \ldots, \lambda_k) \\
&= \sum_{\alpha_1, \ldots, \alpha_k \in \mathbb{F}_q} \prod_{i=1}^{k} \lambda_i(\alpha_i) \\
&= \prod_{i=1}^{k} \left(\sum_{\alpha \in \mathbb{F}_q} \lambda_i(\alpha_i) \right) = 0 && [\text{By } \textit{Theorem } 3.2.5(i)]
\end{aligned}$$

Since λ_k is non-trivial and $\prod_{i=1}^{k} \lambda_i$ is trivial using Lemma 3.4.2 we have

$$\lambda_k(-1) \cdot (q-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1}) + (q-1) J(\lambda_1, \ldots, \lambda_k) = 0 \iff J(\lambda_1, \ldots, \lambda_k) = -\lambda_k(-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1})$$

So we already got the first expression for the second case. Since $\prod_{i=1}^{k-1} \lambda_i$ is non-trivial by induction hypothesis we get

$$\begin{aligned}
\lambda_k(-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1}) &= \lambda_k(-1) \frac{G(\lambda_1, \chi) \cdots G(\lambda_{k-1}, \chi)}{G(\lambda_1 \cdots \lambda_{k-1}, \chi)} \\
&= \lambda_k(-1) \frac{G(\lambda_1, \chi) \cdots G(\lambda_k, \chi)}{G(\overline{\lambda_k}, \chi) \cdot G(\lambda_k, \chi)} && [\text{Since } \lambda_1 \cdots \lambda_k \text{ trivial, } \lambda_1 \cdots \lambda_{k-1} = \overline{\lambda_k}] \\
&= \lambda_k(-1) \frac{G(\lambda_1, \chi) \cdots G(\lambda_k, \chi)}{\lambda_k(-1) q} && [\text{By } \textit{Lemma } 3.3.2(iv)] \\
&= \frac{1}{q} \cdot G(\lambda_1, \chi) \cdots G(\lambda_k, \chi)
\end{aligned}$$

Hence we obtain the second expression. Therefore we have the theorem. ■

Now we have all the results to calculate the absolute value of the Jacobi sum for non-trivial multiplicative characters. So we have the following theorem.

Theorem 3.4.4

Let $\lambda_1, \ldots, \lambda_k \in \boldsymbol{\mathcal{M}}_q$ be non-trivial multiplicative characters of $\mathbb{F}_q$. Then

$$|J(\lambda_1, \ldots, \lambda_k)| = \begin{cases} q^{(k-1)/2} & \text{If } \prod_{i=1}^k \lambda_i \text{ is non-trivial} \\ q^{(k-2)/2} & \text{If } \prod_{i=1}^k \lambda_i \text{ is trivial} \end{cases}$$

Proof: Suppose $\prod_{i=1}^k \lambda_i$ is non-trivial. Then for any non-trivial additive character $\chi \in \boldsymbol{\mathcal{X}}_q$ of $\mathbb{F}_q$ by Theorem 3.4.3 we get

$$J(\lambda_1, \ldots, \lambda_k) = \frac{G(\lambda_1, \chi) \cdot G(\lambda_2, \chi) \cdots G(\lambda_k, \chi)}{G(\lambda_1 \cdots \lambda_k, \chi)}$$

Since each λ_i non-trivial by Theorem 3.3.1 we get $|G(\lambda_i, \chi)| = q^{1/2}$ for all $i \in [k]$ and $|G(\lambda_1 \cdots \lambda_k, \chi)| = q^{1/2}$. Hence we get

$$|J(\lambda_1, \ldots, \lambda_k)| = \frac{\prod_{i=1}^k |G(\lambda_i, \chi)|}{|G(\lambda_1 \cdots \lambda_k, \chi)|} = q^{(q-1)/2}$$

Now suppose $\prod_{i=1}^k \lambda_i$ is trivial. Then again by Theorem 3.4.3 we have

$$J(\lambda_1, \ldots, \lambda_k) = -\lambda_k(-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1})$$

Now since $\prod_{i=1}^k \lambda_i$ is trivial and λ_k is non-trivial we have $\prod_{i=1}^{k-1} \lambda_i$ is non-trivial. Hence by using the first case we have $|J(\lambda_1, \ldots, \lambda_{k-1})| = q^{(k-2)/2}$. Hence $J(\lambda_1, \ldots, \lambda_k)| = q^{(k-2)/2}$. Therefore we have the result. ■

Using the value of $J(\lambda_1, \ldots, \lambda_k)$ we can also get the absolute value of $J_0(\lambda_1, \ldots, \lambda_k)$ when all λ_i's are non-trivial and $\prod_{i=1}^k \lambda_i$ is trivial.

Corollary 3.4.5

Let $\lambda_1, \ldots, \lambda_k \in \boldsymbol{\mathcal{M}}_q$ be non-trivial multiplicative characters of $\mathbb{F}_q$ and $\prod_{i=1}^k \lambda_i$ is trivial. Then

$$|J_0(\lambda_1, \ldots, \lambda_k)| = (q-1)q^{(k-2)/2}$$

Proof: Using Lemma 3.4.2 we have $J_0(\lambda_1, \ldots, \lambda_k) = \lambda_k(-1) \cdot (q-1) \cdot J(\lambda_1, \ldots, \lambda_{k-1})$. Now since $\prod_{i=1}^k \lambda_i$ is trivial and λ_k is non-trivial we have $\prod_{i=1}^{k-1} \lambda_i$ is non-trivial. Therefore by Theorem 3.4.4 we have $|J(\lambda_1, \ldots, \lambda_{k-1})| = q^{(k-2)/2}$. Therefore together we have the corollary. ■

Combining Lemma 3.4.1, Lemma 3.4.2, Theorem 3.4.4, and the corollary above, we obtain a complete description of the absolute values of Jacobi sums in all cases.

Theorem 3.4.6

Let $\lambda_1, \ldots, \lambda_k \in \boldsymbol{\mathcal{M}}_q$ be multiplicative characters of $\mathbb{F}_q$. Then

(i) If λ_i is trivial for all $i \in [k]$ then $J(\lambda_1, \ldots, \lambda_k) = J_0(\lambda_1, \ldots, \lambda_k) = q^{k-1}$.

(ii) If some but not all λ_i's are trivial then $J(\lambda_1, \ldots, \lambda_k) = J_0(\lambda_1, \ldots, \lambda_k) = 0$.

(iii) If all λ_i's are non-trivial and $\prod_{i=1}^{k} \lambda_i$ is non-trivial then $|J(\lambda_1, \ldots, \lambda_k)| = q^{(k-1)/2}$ and $J_0(\lambda_1, \ldots, \lambda_k) = 0$.

(iv) If all λ_i's are non-trivial and $\prod_{i=1}^{k} \lambda_i$ is trivial then $|J(\lambda_1, \ldots, \lambda_k)| = q^{(k-2)/2}$ and $|J_0(\lambda_1, \ldots, \lambda_k)| = (q-1)q^{(k-2)/2}$.

Now the case $k = 2$ often appears in most of the applications of Jacobi sums. Some of those results are of special interest.

Lemma 3.4.7

Let λ be a multiplicative character of $\mathbb{F}_q$ of order $m \geq 2$ and let χ be a non-trivial additive character of $\mathbb{F}_q$. Then

$$G(\lambda, \chi)^m = \lambda(-1) \cdot q \cdot J(\lambda, \lambda) \cdot J(\lambda, \lambda^2) \cdots J(\lambda, \lambda^{m-2})$$

Proof: If $m = 2$ then $\overline{\lambda} = \lambda$ and so $G(\lambda, \chi)^2 = G(\lambda, \chi) \cdot G(\overline{\lambda}, \chi)$ and the right hand side becomes $\lambda(-1) \cdot q$ which we already have from Lemma 3.3.2(iv). So now suppose $m \geq 3$.

Then from Theorem 3.4.3 we have for all $i \in [m-2]$

$$J(\lambda, \lambda^i) = \frac{G(\lambda, \chi) \cdot G(\lambda^i, \chi)}{G(\lambda^{i+1}, \chi)}$$

Therefore taking product of all $J(\lambda, \lambda^i)$ for all $i \in [m-2]$ we get

$$\begin{aligned}
\prod_{i=1}^{m-2} J(\lambda, \lambda^i) &= \prod_{i=1}^{m-2} \frac{G(\lambda, \chi) \cdot G(\lambda^i, \chi)}{G(\lambda^{i+1}, \chi)} \\
&= G(\lambda, \chi)^{m-2} \frac{G(\lambda, \chi)}{G(\lambda^{m-1}, \chi)} \\
&= \frac{G^{m-1}(\lambda, \chi)}{G(\lambda^{m-1}, \chi)}
\end{aligned}$$

Since $\mathrm{ord}(\lambda) = m$ we have $\lambda^{m-1} = \overline{\lambda}$. Therefore we have

$$\prod_{i=1}^{m-2} J(\lambda, \lambda^i) = \frac{G^{m-1}(\lambda, \chi)}{G(\lambda^{m-1}, \chi)} = \frac{G^m(\lambda, \chi)}{G(\overline{\lambda}, \chi) \cdot G(\lambda, \chi)} = \frac{G^m(\lambda, \chi)}{\lambda(-1) \cdot q}$$

Therefore we have the lemma. ■

Now we give another result of relation between Gaussian sums for the $k = 2$ case. This gives relation between two Gaussian sums on non-trivial multiplicative characters of different orders using Jacobi sums. We will not prove this theorem as it is far too complicated to be included in here.

Theorem 3.4.8 Davenport–Hasse Product Formula

Let $\psi, \lambda \in \boldsymbol{\mathcal{M}}_q$ be multiplicative characters of $\mathbb{F}_q$ such that $\mathrm{ord}(\lambda) = m \geq 2$ and ψ^m is non-trivial. Let $\chi_b \in \boldsymbol{\chi}_q$ be a non-trivial additive character of $\mathbb{F}_q$ as defined in Theorem 3.2.1. Then

$$\frac{G(\psi, \chi_b)^m}{G(\psi^m, \chi_{mb})} = \prod_{j=1}^{m-1} J(\psi, \lambda^j).$$

Over the prime field $\mathbb{F}_p$ the Davenport–Hasse relation admits a more explicit form, in which the root of unity appearing on the right hand side is written in terms of a primitive root modulo p. This form appears as Theorem 3 in the paper of Huard, Spearman and Williams.

Theorem 3.4.9 Davenport–Hasse Relation over $\mathbb{F}_p$, [HSW95, Theorem 3]

Let $m \geq 2$ and $n \geq 1$ be integers, let p be an odd prime with $p \equiv 1 \bmod mn$, and set $f = (p-1)/mn$. Let g be a primitive root modulo p, and for any integer $l \not\equiv 0 \bmod p$ let $\operatorname{ind}_g(l)$ denote the least nonnegative integer with $g^{\operatorname{ind}_g(l)} \equiv l \bmod p$. Set $\beta_{mn} = e^{2\pi i/mn}$ and let $\lambda \in \boldsymbol{\mathcal{M}}_p$ be the multiplicative character of order mn determined by $\lambda(g) = \beta_{mn}$. Let $\chi \in \boldsymbol{\mathcal{X}}_p$ be a non-trivial additive character of $\mathbb{F}_p$. Then for every $t = 1, \ldots, m-1$,

$$\frac{G(\lambda^{tn}, \chi) \cdot \prod_{j=1}^{n-1} G(\lambda^{mj}, \chi)}{\prod_{j=0}^{n-1} G(\lambda^{mj+t}, \chi)} = \beta_{mn}^{nt \cdot \operatorname{ind}_g(n)},$$

equivalently,

$$\prod_{j=1}^{n-1} \frac{J(\lambda^{mj}, \lambda^t)}{J(\lambda^{tj}, \lambda^t)} = \beta_{mn}^{nt \cdot \operatorname{ind}_g(n)}.$$

As an application of the Davenport–Hasse product formula, together with the evaluation of Jacobi sums in Theorem 3.4.6, we obtain the following closed form for the product of Gaussian sums on the m characters $\psi \cdot \lambda^i$ twisting ψ by the powers of λ. The two cases reflect the parity of m, with the even case picking up an extra quadratic Gaussian sum $G(\eta, \chi_b)$ where η is the quadratic character of $\mathbb{F}_q$.

Corollary 3.4.10

Let $\psi, \lambda \in \boldsymbol{\mathcal{M}}_q$ be multiplicative characters of $\mathbb{F}_q$ such that $\operatorname{ord}(\lambda) = m \geq 2$ and ψ^m is non-trivial. Let $\chi_b \in \boldsymbol{\mathcal{X}}_q$ be a non-trivial additive character of $\mathbb{F}_q$ as defined in Theorem 3.2.1.

$$\prod_{j=0}^{m} G(\psi \cdot \lambda^j, \chi_b) = \begin{cases} q^{(m-1)/2} G(\psi^m, \chi_{mb}) & \text{if } m \text{ is odd} \\ (-1)^{(q-1)(m-2)/8} \cdot q^{(m-2)/2} \cdot G(\eta, \chi_b) \cdot G(\psi^m, \chi_{mb}) & \text{if } m \text{ is even} \end{cases}$$

Proof: Since $\operatorname{ord}(\lambda) \geq 2$ and ψ^m is non-trivial we have $\psi \cdot \lambda^j$ is non-trivial for all $j \in [m-1]$. By Davenport-Hasse Product Formula we have

$$\frac{G(\psi, \chi_b)^m}{G(\psi^m, \chi_{mb})} = \prod_{j=1}^{m-1} J(\psi, \lambda^j)$$

Now by using Theorem 3.4.3 on the right hand side of this expression we get

$$\frac{G(\psi, \chi_b)^m}{G(\psi^m, \chi_{mb})} = \prod_{j=1}^{m-1} \frac{G(\psi, \chi_b) \cdot G(\lambda^j, \chi_b)}{G(\psi \cdot \lambda^i, \chi_b)} = G(\psi, \chi_b)^{m-1} \prod_{i=1}^{j} \frac{G(\lambda^j, \chi_b)}{G(\psi \cdot \lambda^i, \chi_b)}$$

So together we have

$$\prod_{j=0}^{m-1} G(\psi \cdot \lambda^j, \chi_b) = G(\psi^m, \chi_{mb}) \prod_{j=1}^{m-1} G(\lambda^j, \chi_b) \tag{3.6}$$

Notice we already obtained the left hand side and one term in the right hand side correctly. So now the value at right only depends on the product of $G(\lambda^j, \chi_b)$'s.

Now if m is odd then we have

$$\begin{aligned}
\prod_{j=1}^{m-1} G(\lambda^j, \chi_b) &= \prod_{j=1}^{(m-1)/2} G(\lambda^j, \chi_b) G(\lambda^{m-j}, \chi_b) \\
&= \prod_{j=1}^{(m-1)/2} G(\lambda^j, \chi_b) \cdot G(\overline{\lambda^j}, \chi_b) && [\text{As } \operatorname{ord}(\lambda) = m = \text{odd}] \\
&= \prod_{j=1}^{(m-1)/2} \lambda^j(-1) \cdot q && [\text{By } \textit{Lemma } 3.3.2(iv)] \\
&= q^{(m-1)/2} \prod_{j=1}^{(m-1)/2} \lambda^j(-1)
\end{aligned}$$

Since m is odd by Observation 3.6 we have $\lambda(-1) = 1$. Therefore

$$\prod_{j=1}^{m-1} G(\lambda^j, \chi_b) = q^{(m-1)/2}$$

So we have the first case together with (3.6). Now for the remaining case of m being even we have

$$\begin{aligned}
\prod_{j=1}^{m-1} G(\lambda^j, \chi_b) &= G(\lambda^{m/2}, \chi_b) \prod_{j=1}^{(m-2)/2} G(\lambda^j, \chi_b) G(\lambda^{m-j}, \chi_b) \\
&= G(\eta, \chi_b) \prod_{j=1}^{(m-2)/2} G(\lambda^j, \chi_b) \cdot G(\overline{\lambda^j}, \chi_b) && [\text{As } \operatorname{ord}(\lambda) = m = \text{even}] \\
&= G(\eta, \chi_b) \prod_{j=1}^{(m-2)/2} \lambda^j(-1) \cdot q && [\text{By } \textit{Lemma } 3.3.2(iv)] \\
&= q^{(m-2)/2} \cdot G(\eta, \chi_b) \prod_{j=1}^{(m-2)/2} \lambda^j(-1)
\end{aligned}$$

Again by Observation 3.6 since m is even if $\frac{q-1}{m}$ is odd then $\lambda(-1) = 1$ and if $\frac{q-1}{m}$ is even then $\lambda(-1) = 1$. Hence we can write $\lambda(-1) = (-1)^{(q-1)/m}$. Therefore

$$\prod_{j=1}^{m-1} G(\lambda^j, \chi_b) = (-1)^{(q-1)(m-2)/8} \cdot q^{(m-2)/2} \cdot G(\eta, \chi_b)$$

So together with (3.6) we have the result. ■

A classical application of the arithmetic of Jacobi sums is a short representation-theoretic proof of a celebrated theorem of Fermat: every prime $p \equiv 1 \bmod 4$ is a sum of two integer squares. The key point is that when λ is a multiplicative character of $\mathbb{F}_p$ of order 4, the Jacobi sum $J(\lambda, \eta)$ lies in the ring of Gaussian integers $\mathbb{Z}[i]$ and has absolute value exactly $\sqrt{p}$.

Theorem 3.4.11 Fermat's Two-Square Theorem

Let p be an odd prime with $p \equiv 1 \bmod 4$, let η denote the quadratic character of $\mathbb{F}_p$, and let $\lambda \in \mathcal{M}_p$ be a multiplicative character of $\mathbb{F}_p$ of order 4. Then there exist integers $A, B \in \mathbb{Z}$ such that

$$J(\lambda, \eta) = A + iB \qquad \textit{and} \qquad p = A^2 + B^2.$$

Proof: Since $\mathrm{ord}(\lambda) = 4$, λ takes values in $\{\pm 1, \pm i\} \subseteq \mathbb{Z}[i]$, and since η takes values in $\{\pm 1\} \subseteq \mathbb{Z}$, the Jacobi sum

$$J(\lambda, \eta) = \sum_{\alpha \in \mathbb{F}_p} \lambda(\alpha) \cdot \eta(1-\alpha) \in \mathbb{Z}[i].$$

Write $J(\lambda, \eta) = A + iB$ with $A, B \in \mathbb{Z}$. Since λ and η are both non-trivial and $\lambda \cdot \eta = \lambda^3$ is also non-trivial (as $\mathrm{ord}(\lambda) = 4$), Theorem 3.4.4 gives $|J(\lambda, \eta)| = p^{1/2}$; equivalently $A^2 + B^2 = p$. ■

Corollary 3.4.12

Let $\psi, \lambda \in \boldsymbol{\mathcal{M}}_q$ be multiplicative characters of $\mathbb{F}_q$ such that $\mathrm{ord}(\lambda) = m \geq 2$ and ψ^m is non-trivial. Let $\chi \in \boldsymbol{\mathcal{X}}_q$ be any non-trivial additive character of $\mathbb{F}_q$. Then

$$\prod_{j=0}^{m-1} G(\psi \cdot \lambda^j) = \psi^{-m}(m) \cdot G(\psi^m, \chi) \prod_{j=1}^{m-1} G(\lambda^j, \chi)$$

where for any $\gamma \in \mathbb{F}_q$, $\chi_m(\gamma) = \chi(m \cdot \gamma)$ similar to as defined in Theorem 3.2.1.

Proof: By Davenport-Hasse Product Formula we have

$$\frac{G(\psi, \chi)^m}{G(\psi^m, \chi)} = \prod_{j=1}^{m} J(\psi, \lambda^j)$$

Now using Theorem 3.4.3 for any $j \in \{m-1\}$ we have

$$J(\psi, \lambda^j) = \frac{G(\psi, \chi) \cdot G(\lambda^j, \chi}{G(\psi \cdot \lambda^j, \chi)}$$

So replacing $J(\psi, \lambda^j)$ we get

$$\frac{G(\psi, \chi)^m}{G(\psi^m, \chi)} = \prod_{j=1}^{m-1} \frac{G(\psi, \chi) \cdot G(\lambda^j, \chi}{G(\psi \cdot \lambda^j, \chi)} = G(\psi, \chi)^{m-1} \prod_{j=1}^{m-1} \frac{G(\lambda^j, \chi}{G(\psi \cdot \lambda^j, \chi)}$$

So we get the final relation

$$\prod_{j=0}^{m-1} G(\psi \cdot \lambda^j) = G(\psi^m, \chi_m) \prod_{j=1}^{m-1} G(\lambda^j, \chi) \tag{3.7}$$

Now by Lemma 3.3.2(i) we have $G(\psi^m, \chi_m) = \psi^{-m}(m) G(\psi^m, \chi)$. So replacing $G(\psi^m, \chi_m)$ we get the theorem. ■

Since for the trivial multiplicative character $\psi_0 \in \boldsymbol{\mathcal{M}}_q$ we have $\lambda^0 = \psi_0$. Therefore $\prod_{j=0}^{m-1} G(\lambda^j, \chi) = G(\psi_0, \chi) \prod_{j=1}^{m-1} G(\lambda^j, \chi) = -\prod_{j=1}^{m-1} G(\lambda^j, \chi)$ by Theorem 3.3.1. Therefore from (3.7) we have

$$\prod_{j=0}^{m-1} G(\psi \cdot \lambda^j) = -G(\psi^m, \chi_m) \prod_{j=0}^{m-1} G(\lambda^j, \chi)$$

So we get the following observation:

Observation 3.11. *Let $\psi, \lambda \in \boldsymbol{\mathcal{M}}_q$ be multiplicative characters of $\mathbb{F}_q$ such that $\mathrm{ord}(\lambda) = m \geq 2$ and ψ^m is non-trivial. Let $\chi \in \boldsymbol{\mathcal{X}}_q$ be any non-trivial additive character of $\mathbb{F}_q$. Then*

$$\prod_{j=0}^{m-1} G(\psi \cdot \lambda^j) = -\psi^{-m}(m) \cdot G(\psi^m, \chi) \prod_{j=0}^{m-1} G(\lambda^j, \chi)$$

3.5 L-functions

The characters of cyclic groups defined in Lemma 3.1.1 are precisely the classical *Dirichlet characters*. Dirichlet characters are of paramount importance in analytic number theory: they appear in the study of primes in arithmetic progressions, the distribution of prime ideals, and the analytic continuation of L-functions. In the finite-field setting, the multiplicative characters of $\mathbb{F}_q$ play an analogous role, and the associated L-functions encode deep arithmetic information about character sums over $\mathbb{F}_q$.

Before specialising to the function-field setting, it is helpful to pause and describe the general analytic framework into which all of these objects fit. The central analytic device is a *Dirichlet series*

$$D(s) = \sum_{n=1}^{\infty} \frac{a_n}{n^s}, \qquad s \in \mathbb{C},$$

a formal packaging of an arithmetic sequence $(a_n)_{n\geq 1}$ into a single complex-analytic function. The key insight, going back to Euler and Dirichlet, is that analytic properties of $D(s)$ — its region of convergence, its poles, its zeros, its functional equation — encode deep arithmetic information about the coefficients a_n. Three families of Dirichlet series organise most of analytic number theory, arranged in a nested hierarchy:

1. **Zeta functions** arise when the coefficients are the trivial ones, $a_n \equiv 1$, so that the Dirichlet series simply counts the size of some arithmetic object. The prototype is the *Riemann zeta function* $\zeta(s) = \sum_n n^{-s}$, which encodes the multiplicative structure of $\mathbb{Z}$. More generally, any reasonable "arithmetic scheme" — the ring of integers of a number field, the polynomial ring over a finite field, an algebraic variety — has its own zeta function built in the same way.

2. L**-functions** arise when the coefficients are *twisted* by a character χ, giving $L(s,\chi) = \sum_n \chi(n)/n^s$. The character χ is what allows one to isolate arithmetic information that a zeta function averages over. Where ζ sees only the coarse fact that $\mathbb{Z}$ is generated by primes, $L(s,\chi)$ sees how those primes distribute themselves across residue classes modulo some conductor — which is exactly the input Dirichlet used to prove the infinitude of primes in arithmetic progressions. Taking the trivial character $\chi \equiv 1$ recovers ζ up to finitely many Euler factors, so zeta functions are the "character-free" special case of L-functions.

3. **General Dirichlet series** with coefficients attached to richer arithmetic objects: Fourier coefficients of modular forms, traces of Frobenius on Galois representations, Hecke eigenvalues, and so on. These have fewer a priori properties, but conjecturally (the Langlands programme) every "motivic" Dirichlet series is itself an L-function of some automorphic representation.

The unifying phenomenon across all three families is a single package of analytic properties: absolute convergence in some right half-plane $\Re(s) > \sigma_0$; an *Euler product* $\prod_p E_p(p^{-s})^{-1}$ expressing $D(s)$ multiplicatively over primes — the analytic avatar of unique factorisation; a meromorphic continuation to all of $\mathbb{C}$; and a *functional equation* relating $D(s)$ to $D(1-s)$ (after multiplication by suitable archimedean gamma factors), reflecting a hidden duality. The locations of poles and zeros of $D(s)$ then govern arithmetic counting problems: the pole gives a main term (as in the Prime Number Theorem), while the zeros give the oscillating corrections.

The rest of this section unfolds this dictionary in two parallel worlds. First we recall the classical story over $\mathbb{Q}$ — the Riemann zeta function, its zeros, and its L-function generalisations — so that the relevant analytic phenomena are in view. We then develop the function-field analog: the zeta function associated to $\mathbb{F}_q[X]$ and its twists by multiplicative characters of $\mathbb{F}_q$. The analogy is strikingly faithful, but the function-field side is in many ways easier — the zeta function turns out to be a rational function of q^{-s}, and the analog of the Riemann Hypothesis is a theorem (due to Weil) rather than an open problem.

3.5.1 The Classical Riemann Zeta Function

The prototype of everything that follows is the *Riemann zeta function*

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}, \qquad \Re(s) > 1,$$

which converges absolutely in the right half-plane $\Re(s) > 1$. Most proofs in this subsection are omitted, especially those relying on contour integration or the Mellin transform; the emphasis is on statements together with their arithmetic meaning.

Theorem 3.5.1 Euler Product

For $\Re(s) > 1$,

$$\zeta(s) = \prod_{p \text{ prime}} \frac{1}{1 - p^{-s}}.$$

The identity is the analytic translation of *unique factorisation*: expanding each geometric factor $(1 - p^{-s})^{-1} = \sum_{k \geq 0} p^{-ks}$ and multiplying reproduces the sum over positive integers. Since $\zeta(s)$ has a pole at $s = 1$ (the harmonic series diverges), the product must have infinitely many factors — giving an analytic proof of the infinitude of primes. The same identity is the engine behind more refined statements: taking logarithmic derivatives of both sides and comparing coefficients converts questions about primes into questions about $\zeta(s)$, as we shall see repeatedly below.

Observation 3.12 (Special values)**.** *For every* $n \geq 1$ *there is a rational* $r(n) \in \mathbb{Q}$ *with* $\zeta(2n) = r(n)\,\pi^{2n}$*; in particular*

$$\zeta(2) = \frac{\pi^2}{6}, \qquad \zeta(4) = \frac{\pi^4}{90}.$$

By analytic continuation,

$$\zeta(0) = -\frac{1}{2}, \qquad \zeta(-1) = -\frac{1}{12}.$$

The rationals $r(n)$ are expressible in terms of Bernoulli numbers, tying the special values of ζ at positive even integers to deep algebraic invariants. Odd values are much harder. Apéry (1979) proved that $\zeta(3)$ is irrational; the arithmetic nature of $\zeta(5), \zeta(7), \ldots$ is still unknown. The negative-integer values $\zeta(0), \zeta(-1), \ldots$ make sense only *after* analytic continuation, and are initially surprising: naively $\zeta(-1) = 1 + 2 + 3 + \cdots$, whereas the continuation assigns it the value $-\frac{1}{12}$.

Theorem 3.5.2 Functional Equation

Define the completed zeta function

$$\xi(s) = \tfrac{1}{2}\, s(s-1)\, \Gamma(s/2)\, \pi^{-s/2}\, \zeta(s).$$

Then ξ *extends to an entire function on* $\mathbb{C}$ *and satisfies*

$$\xi(s) = \xi(1 - s).$$

The factors $s(s-1)$ are there to kill off two unwanted poles: the simple pole of ζ at $s = 1$, and the pole that $\Gamma(s/2)$ introduces at $s = 0$. The remaining factor $\Gamma(s/2)\pi^{-s/2}$ plays the role of an "archimedean Euler factor" — a contribution from the prime at infinity that completes the Euler product into a fully symmetric object. The proof (omitted) proceeds via the Mellin transform of the theta function $\theta(x) = \sum_{n \in \mathbb{Z}} e^{-\pi n^2 x}$, which via Poisson summation satisfies the modular transformation $\theta(1/x) = \sqrt{x}\,\theta(x)$. In this sense the symmetry $s \leftrightarrow 1 - s$ of ζ is really the modular transformation of θ in disguise. As a consequence, $\zeta(s)$ admits a meromorphic continuation to all of $\mathbb{C}$, with a unique simple pole at $s = 1$ of residue 1.

3.5.2 Zeros of $\zeta(s)$ and the Riemann Hypothesis

From the functional equation and the poles of $\Gamma(s/2)$ at $s = 0, -2, -4, \ldots$, one reads off the *trivial zeros* of ζ: the negative even integers $s = -2, -4, -6, \ldots$. All other zeros lie in the *critical strip* $0 \le \Re(s) \le 1$; the functional equation forces them to be symmetric under $s \mapsto 1 - s$, and conjugation $s \mapsto \bar{s}$ forces reflection symmetry across the real axis. The line $\Re(s) = 1/2$ is the *critical line*, and the most famous open problem in mathematics is:

Riemann Hypothesis. *Every non-trivial zero of $\zeta(s)$ satisfies $\Re(s) = 1/2$.*

Theorem 3.5.3 Zero counting

Let $N(T)$ count the zeros $\rho = \beta + i\gamma$ of ζ with $0 < \gamma \le T$. Then

$$N(T) = \frac{T}{2\pi} \log \frac{T}{2\pi} - \frac{T}{2\pi} + O(\log T).$$

So the zeros become logarithmically denser as one moves up the critical strip: near height T their density is $\frac{1}{2\pi} \log(T/2\pi)$. Numerical work has verified RH for the first 10^{13} or so zeros, yet a proof remains elusive.

Theorem 3.5.4 Hadamard Product

There exist constants $A, B \in \mathbb{C}$ such that

$$\xi(s) = e^{A+Bs} \prod_{\rho} \left(1 - \frac{s}{\rho}\right) e^{s/\rho},$$

where the product runs over all non-trivial zeros ρ of ζ.

This is the "zero-side" factorisation of ξ, to be contrasted with the Euler product as the "prime-side" factorisation. Together they are the source of the famous duality between primes and zeros: equating the two factorisations — typically after taking logarithmic derivatives — converts statements about primes into statements about zeros and vice versa.

Zeros $\leftrightarrow$ primes. Taking the logarithmic derivative of the Euler product,

$$-\frac{\zeta'(s)}{\zeta(s)} = \sum_{n=1}^{\infty} \frac{\Lambda(n)}{n^s},$$

where $\Lambda(n) = \log p$ if $n = p^k$ and 0 otherwise (the *von Mangoldt function*). Contour-integrating this Dirichlet series against x^s/s produces the prime-counting sum $\psi(x) = \sum_{n \le x} \Lambda(n)$, while picking up residues at every pole of ζ'/ζ — that is, at $s = 1$ (where ζ has a pole) and at every non-trivial zero ρ (where ζ vanishes). The result is one of the most striking identities in all of mathematics:

Theorem 3.5.5 Riemann–von Mangoldt Explicit Formula

For $x > 1$ not a prime power,

$$\psi(x) = x - \sum_{\rho} \frac{x^\rho}{\rho} - \frac{\zeta'(0)}{\zeta(0)} - \tfrac{1}{2} \log(1 - x^{-2}).$$

The main term x comes from the pole of ζ at $s = 1$ — this *is* the Prime Number Theorem, in sharp form. The sum over ρ is an oscillating correction whose size is controlled by $\Re(\rho)$: each zero $\rho = \beta + i\gamma$ contributes a wave of amplitude x^β and frequency $\gamma/(2\pi)$ on a logarithmic scale. Under RH, each $|x^\rho| = x^{1/2}$, giving

$$\psi(x) = x + O(x^{1/2} \log^2 x),$$

the sharpest conceivable error term for prime counting. The explicit formula thus makes precise the intuition that the primes "know about" the zeros of ζ and vice versa.

PNT $\Leftrightarrow$ non-vanishing on $\Re(s) = 1$. The Prime Number Theorem (in its crude form $\psi(x) \sim x$) is equivalent to the non-vanishing $\zeta(1+it) \neq 0$ for every $t \in \mathbb{R}$. Mertens' elementary trick for proving this uses the inequality

$$3 + 4\cos\theta + \cos 2\theta = 2(1+\cos\theta)^2 \geq 0,$$

which combined with the Euler product forces

$$\zeta(\sigma)^3\,|\zeta(\sigma+it)|^4\,|\zeta(\sigma+2it)| \geq 1 \qquad (\sigma > 1).$$

If $\zeta(1+it_0) = 0$ for some $t_0 \neq 0$, then as $\sigma \to 1^+$ the fourth-order vanishing $|\zeta(\sigma+it_0)|^4 \to 0$ would beat the triple pole $\zeta(\sigma)^3 \sim (\sigma-1)^{-3}$, contradicting the bound ≥ 1. So the pole of ζ at $s = 1$ and the non-vanishing on the line $\Re(s) = 1$ are two sides of the same coin.

Theorem 3.5.6 Hardy, 1914

Infinitely many non-trivial zeros of $\zeta(s)$ lie on the critical line $\Re(s) = 1/2$.

Hardy's result is unconditional, but does not prove RH — only that RH is not violated "everywhere." Later refinements (Selberg, Levinson, Conrey) show that a positive proportion — currently at least 41% — of the non-trivial zeros lie on the critical line.

The Riemann–Siegel Ξ-function. Setting $s = \frac{1}{2} + it$ and $\Xi(t) = \xi(\frac{1}{2} + it)$, the functional equation $\xi(s) = \xi(1-s)$ becomes $\Xi(-t) = \Xi(t)$, and Ξ is a real-valued, even, entire function of t. The Riemann Hypothesis then becomes the assertion that *all* zeros of Ξ are real; Hardy's theorem says Ξ has infinitely many real zeros. This is the form in which RH is most often attacked numerically.

Theorem 3.5.7 Li's Criterion

Define

$$\lambda_n = \sum_{\rho} \left[1 - \left(1 - \frac{1}{\rho}\right)^n\right], \qquad n \geq 1,$$

the sum running over all non-trivial zeros of ζ. Then

$$\text{RH} \iff \lambda_n > 0 \textit{ for every } n \geq 1.$$

Voros's asymptotic reformulation: assuming RH,

$$\lambda_n \sim \tfrac{1}{2}\, n\big(\log n - \log 2\pi - 1 + \gamma\big),$$

where γ is Euler's constant. So RH is equivalent to a single sequence of real numbers being positive and growing like $\frac{1}{2} n \log n$ — converting a statement about zero locations into a statement about positivity of an explicit sequence. Attractive, but no easier to prove.

Generalised Riemann Hypothesis. Each Dirichlet character χ mod q gives rise to an L-function

$$L(s,\chi) = \sum_{n=1}^{\infty} \frac{\chi(n)}{n^s} = \prod_p \frac{1}{1 - \chi(p)p^{-s}},$$

which enjoys a meromorphic continuation and a functional equation relating $s \leftrightarrow 1-s$. The *Generalised Riemann Hypothesis* asserts that all non-trivial zeros of every such $L(s,\chi)$ lie on $\Re(s) = 1/2$. The non-vanishing $L(1,\chi) \neq 0$ for

non-principal χ is exactly the input Dirichlet needed for his theorem on primes in arithmetic progressions — the prototype "L-function application."

3.5.3 Zeta Functions over $\mathbb{F}_q$

The function-field world has its own zeta function, obtained by replacing $\mathbb{Z}$ with the polynomial ring $\mathbb{F}_q[X]$ and the primes with monic irreducible polynomials. The notational parallel with the classical case is exact, but the analytic behaviour is dramatically simpler: the resulting zeta function is a *rational* function of q^{-s}, with no mystery in its analytic continuation, and (as will be developed in later sections) a fully-proven analog of the Riemann Hypothesis.

Let $\Phi \subseteq \mathbb{F}_q[X]$ denote the set of all monic polynomials, and let $\Phi_d \subseteq \Phi$ be the subset of polynomials of degree exactly d, for each $d \in \mathbb{N}$. A basic count gives $|\Phi_d| = q^d$: a degree-d monic polynomial is determined by its d lower-order coefficients, each of which is a free element of $\mathbb{F}_q$. This simple combinatorial fact is what makes the function-field zeta function so much more tractable than its classical counterpart.

For any polynomial $f \in \mathbb{F}_q[X]$, the quotient ring $\mathbb{F}_q[X] \,/\, \langle f \rangle$ is a finite $\mathbb{F}_q$-algebra of size $q^{\deg(f)}$. We define the *norm* of f to be

$$\mathfrak{n}(f) = \left| \mathbb{F}_q[X] \Big/ \langle f \rangle \right| = q^{\deg(f)},$$

by direct analogy with the norm of an ideal in the ring of integers of a number field. This norm is the direct replacement for the size of $\mathbb{Z}/n\mathbb{Z}$ (namely n) that appears in the classical Riemann zeta function.

Definition 3.5.1: Zeta Function

For any complex number $s = a + i \cdot b \in \mathbb{C}$ the zeta function *of $\mathbb{F}_q[X]$ at s is*

$$\zeta(s) = \sum_{h \in \Phi} \frac{1}{\mathfrak{n}(h)^s},$$

where Φ is the set of all monic polynomials over $\mathbb{F}_q$.

Grouping the terms of this series by degree and using $|\Phi_d| = q^d$ gives a closed-form evaluation:

$$\zeta(s) = \sum_{h \in \Phi} \frac{1}{\mathfrak{n}(h)^s} = \sum_{d=0}^{\infty} \frac{|\Phi_d|}{q^{d \cdot s}} = \sum_{d=0}^{\infty} \frac{q^d}{q^{d \cdot s}} = \sum_{d=0}^{\infty} q^{d(1-s)}.$$

The rightmost sum is a geometric series; it is absolutely convergent for $\Re(s) > 1$ and uniformly convergent on every half-plane $\Re(s) \geq a_0 > 1$, and it sums to an explicit rational function.

Observation 3.13. *For any $s = a + i \cdot b \in \mathbb{C}$ with $\Re(s) > 1$,*

$$\zeta(s) = \sum_{d=0}^{\infty} q^{d(1-s)} = \frac{1}{1 - q^{1-s}}.$$

This closed form is already a striking contrast with the classical Riemann zeta function, whose analytic continuation and functional equation occupy pages of proof. Here $\zeta(s) = (1 - q^{1-s})^{-1}$ extends immediately to a meromorphic function on all of $\mathbb{C}$, with simple poles at the points where $q^{1-s} = 1$ (i.e. $s = 1 + \frac{2\pi i k}{\log q}$ for $k \in \mathbb{Z}$) and no zeros whatsoever. The elementary reason for this collapse of difficulty is the exact combinatorial identity $|\Phi_d| = q^d$; counting integers by size has no such structure.

Just as the classical Riemann zeta function admits an Euler product over the primes, the zeta function over $\mathbb{F}_q$ admits an analogous product over the monic irreducible polynomials in $\mathbb{F}_q[X]$ — the "primes" of the polynomial ring.

Lemma 3.5.8

Let $s = a + i \cdot b \in \mathbb{C}$ with $\Re(s) > 1$. Then

$$\zeta(s) = \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} (1 - \mathfrak{n}(h)^{-s})^{-1}.$$

Proof: Using unique factorisation of $\mathbb{F}_q[X]$ into monic irreducibles, we have

$$\begin{aligned}
\zeta(s) &= \sum_{h \in \Phi} \frac{1}{\mathfrak{n}(h)^s} \\
&= \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} \left(1 + \frac{1}{\mathfrak{n}(h)^s} + \frac{1}{\mathfrak{n}(h^2)^s} + \cdots\right) \\
&= \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} \left(\sum_{i=0}^{\infty} \frac{1}{\mathfrak{n}(h^i)^s}\right) \\
&= \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} \left(\sum_{i=0}^{\infty} \frac{1}{\mathfrak{n}(h)^{i \cdot s}}\right) \\
&= \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} (1 - \mathfrak{n}(h)^{-s})^{-1}.
\end{aligned}$$

So we have the lemma. ■

Equating the closed form $\zeta(s) = (1 - q^{1-s})^{-1}$ with the Euler product gives a function-field analog of the Euler identity:

$$\frac{1}{1 - q^{1-s}} = \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} (1 - \mathfrak{n}(h)^{-s})^{-1}.$$

Comparing Dirichlet-series coefficients on the two sides leads directly to an exact count of the number of monic irreducible polynomials of each degree — a function-field "prime polynomial theorem" which, unlike its classical counterpart, is elementary and exact rather than asymptotic.

3.5.4 Dirichlet L-function

To define L-functions in the function field setting we first need to extend multiplicative characters to the polynomial ring. Let $G \subseteq \mathbb{F}_q(X)$ be the group of rational functions $h_1(X)/h_2(X)$ where $h_2 \neq 0$ and $h_1, h_2 \in \Phi$ where $\Phi \subseteq G$ is the set of all monic polynomials over $\mathbb{F}_q$. From now on we use $h(X) \in G$ to denote monic polynomials. Let $\overline{G}$ be the subgroup of G such that $h_1 \cdot h_2 \in \overline{G}$ whenever $h_1, h_2 \in \overline{G}$. For a character χ on $\overline{G}$ we extend χ to all of G by setting $\chi(r) = 0$ for all $r \notin \overline{G}$, after which χ remains multiplicative on G. This allows us to sum χ over all monic polynomials.

Definition 3.5.2: Dirichlet L-function

Let χ be a character on $\overline{G}$ extended to G as above. For a complex number $s = a + ib \in \mathbb{C}$ with $a > 1$, the Dirichlet L-function *associated to χ is defined by*

$$L(s, \chi) = \sum_{h \in \Phi} \chi(h) \cdot \mathfrak{n}(h)^{-s}$$

where the sum runs over all monic polynomials $h \in \Phi \subseteq G$ and $\mathfrak{n}(h) = q^{\deg(h)}$ is the norm of h.

Again this sum is absolutely convergent for $\Re(s) > 1$. Like in the case of Riemann-Zeta functions we can express $L(s, \chi)$ in a Euler product form using only irreducible polynomials.

Theorem 3.5.9

For $\Re(s) > 1$ we have

$$L(s, \chi) = \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} (1 - \chi(h) \cdot \mathfrak{n}(h)^{-s})^{-1}$$

Proof: Since $\Re(s) > 1$ we have

$$L(s, \chi) = \sum_{h \in \Phi} \chi(h) \cdot \mathfrak{n}(h)^{-s} = \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} \left(\sum_{i=0}^{\infty} \chi(h^i) \cdot \mathfrak{n}(h^i)^{-s} \right) = \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} (1 - \chi(h) \cdot \mathfrak{n}(h)^{-s})^{-1}$$

Hence we have the lemma. ■

So let $\chi : G \to \mathbb{C}$ be a multiplicative complex valued function over the G where G is as defined above which also satisfies $|\chi(r)| \leq 1$ for all $r \in G$ and $\chi(1) = 1$. Then for any $s \in \mathbb{C}$ with $\Re(s) > 1$ we can rewrite

$$L(s, \chi) = \sum_{h \in \Phi} \chi(h) \cdot \mathfrak{n}(h)^{-s} = \sum_{d=0}^{\infty} \left(\sum_{h \in \Phi_d} \chi(h) \right) q^{-d \cdot s}$$

Since for any $h \in \Phi$ we have $\mathfrak{n}(q)^s = (q^s)^{\deg(h)}$ we can consider the corresponding power series

$$\overline{L}(z, \chi) = \sum_{h \in \Phi} \chi(h) \cdot z^{\deg(h)}$$

hence $\overline{L}(q^{-s}, \chi) = L(s, \chi)$. Now using Theorem 3.5.9 for $\overline{L}(z, \chi)$ we have

$$\overline{L}(z, \chi) = \prod_{\substack{h \in \Phi \\ h \text{ irreducible}}} \left(1 - \chi(h) \cdot z^{\deg(h)}\right)^{-1}$$

Now we do some operations on the power series $\overline{L}(z, \chi)$ to get some expressions which help us later in proving some properties of exponential character sums. So for brevity of notations we use $\mathrm{Irr}(\Phi)$ to denote the set of irreducible polynomials of Φ. We have $\log \overline{L}(z, \chi) = -\sum_{h \in \mathrm{Irr}(\Phi)} \log\left(1 - \chi(h) z^{\deg(h)}\right)$. So by differentiating we get

$$\frac{d}{dz} \log \overline{L}(z, \chi) = -\sum_{h \in \mathrm{Irr}(\Phi)} \frac{-\chi(h) \cdot \deg(h) \cdot z^{\deg(h)-1}}{1 - \chi(h) \cdot z^{\deg(h)}} = \sum_{h \in \mathrm{Irr}(\Phi)} \frac{\chi(h) \cdot \deg(h) \cdot z^{\deg(h)-1}}{1 - \chi(h) \cdot z^{\deg(h)}}$$

So

$$\begin{aligned} z \cdot \frac{d}{dz} \log \overline{L}(z, \chi) &= \sum_{h \in \mathrm{Irr}(\Phi)} \frac{\chi(h) \cdot \deg(h) \cdot z^{\deg(h)-1}}{1 - \chi(h) \cdot z^{\deg(h)}} \\ &= \sum_{h \in \mathrm{Irr}(\Phi)} \chi(h) \cdot \deg(h) \cdot z^{\deg(h)} \sum_{k=0}^{\infty} \chi^k(h) \cdot z^{k \cdot \deg(h)} \\ &= \sum_{h \in \mathrm{Irr}(\Phi)} \deg(h) \sum_{k=1}^{\infty} \chi^k(h) \cdot z^{k \cdot \deg(h)} \\ &= \sum_{k=1}^{\infty} \left(\sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid k}} \deg(h) \cdot \chi(h)^{k/\deg(h)} \right) z^k \end{aligned} \tag{3.8}$$

So define $L_k := \sum_{h\in\mathrm{Irr}(\Phi),\ \deg(h)|k} \deg(h)\cdot\chi(h)^{k/\deg(h)}$. Hence we have $z\cdot\frac{d}{dz}\log\overline{L}(z,\chi) = \sum_{k=1}^{\infty} L_k\cdot z^k$.

Now suppose there exists a positive integer $N\in\mathbb{N}$ such that for all $d>N$ we have $\sum_{h\in\Phi_d}\chi(g)=0$. Then we have

$$L(s,\chi) = \sum_{d=0}^{N}\left(\sum_{h\in\Phi_d}\chi(d)\right)q^{-d\cdot s}, \qquad \overline{L}(z,\chi) = \sum_{d=0}^{N}\left(\sum_{h\in\Phi_d}\chi(d)\right)z^d$$

Since now $\overline{L}(z,\chi)$ is not an infinite power series anymore, instead a degree N polynomial over $\mathbb{C}$ there exists complex numbers $w_1,\cdots,w_N\in\mathbb{C}$ such that $\overline{L}(z,\chi) = (1-w_1z)(1-w_2z)\cdots(1-w_Nz)$. Now applying $z\cdot\frac{d}{dz}\log\overline{L}(z,\chi)$ on this expression we get

$$\begin{aligned} z\cdot\frac{d}{dz}\log\overline{L}(z,\chi) &= -\sum_{i=1}^{N}\frac{w_iz}{1-w_iz} \\ &= -\sum_{i=1}^{N} w_iz\left(\sum_{j=0}^{\infty} w_i^jz^j\right) \\ &= -\sum_{i=1}^{N}\left(\sum_{j=1}^{\infty} w_i^j\right)z^i \end{aligned} \tag{3.9}$$

Therefore by comparing coefficients of (3.9) with (3.8) we get $L_k = -\sum_{i=1}^{N} w_i^k$. for all $k\in\mathbb{N}$. So we have the following theorem

Theorem 3.5.10

Let $\chi : G\to\mathbb{C}$ be a multiplicative function with $|\chi(r)|\le 1$ for all $r\in G$ and $\chi(1)=1$, and suppose there exists a positive integer $N\in\mathbb{N}$ such that $\sum_{h\in\Phi_d}\chi(h)=0$ for every $d>N$. Then $\overline{L}(z,\chi)$ is a polynomial of degree at most N over $\mathbb{C}$, and there exist complex numbers $w_1,\ldots,w_N\in\mathbb{C}$ such that

$$\overline{L}(z,\chi) = \prod_{i=1}^{N}(1-w_iz), \qquad \textit{and} \qquad L(s,\chi) = \prod_{i=1}^{N}\left(1-w_iq^{-s}\right).$$

Moreover, for every $k\in\mathbb{N}$,

$$L_k = \sum_{\substack{h\in\mathrm{Irr}(\Phi)\\ \deg(h)|k}} \deg(h)\cdot\chi(h)^{k/\deg(h)} = -\sum_{i=1}^{N} w_i^k.$$

3.6 Lifting of Gaussian and Jacobi Sums

So using the discussion on *Dirichlet L-functions* we can relate Gaussian sums in $\mathbb{F}_q$ with Gaussian sum of lifted characters to any finite extension of $\mathbb{F}_q$.

Theorem 3.6.1 Davenport-Hasse Theorem

Let $\chi\in\boldsymbol{\mathcal{X}}_q$ be an additive character and $\psi\in\boldsymbol{\mathcal{M}}_q$ be a multiplicative character of $\mathbb{F}_q$, not both of them trivial. Suppose χ and ψ are lifted to characters $\chi^{(r)}$ and $\psi^{(r)}$ respectively of the finite extension field E of $\mathbb{F}_q$ with $[E\colon\mathbb{F}_q]=r$. Then

$$G(\psi^{(r)},\chi^{(r)}) = (-1)^{r-1}G(\psi,\chi)^r$$

Proof: Like previous section G is the set of all rational functions $r(X) = {}^{h_1}/_{h_2}$ where $h_1,h_2\in\Phi$, h_2 non-zero. Let $\overline{G}$ be the subgroup of G consisting of rational functions $r(X) = \frac{h_1(X)}{h_2(X)}$ having $h_1(0),h_2(0)\neq 0$. Then define the multiplicative function $\lambda : G\to\mathbb{C}$ such that for all $r\in G$, $|\lambda(r)|\le 1$ and $\lambda(1)=1$ in the following way:

If $r(X) \in G$ where $r(X) = \frac{h_1(X)}{h_2(X)}$ and in the common splitting field E of h_1, h_2, they factor as:

$$h_1(X) = \prod_{i=1}^{d_1}(X - \alpha_i) \qquad \text{and} \qquad h_1(X) = \prod_{j=1}^{d_2}(X - \beta_j)$$

where $\alpha_i, \beta_j \in E$ for all $i \in [d_1]$, $j \in [d_2]$ then

$$\lambda(r) = \psi\left(\prod_{i=1}^{d_1}\alpha_i \cdot \prod_{j=1}^{d_2}\beta_j^{-1}\right) \cdot \chi\left(\sum_{i=1}^{d_1}\alpha_i - \sum_{j=1}^{d_2}\beta_j\right)$$

Now for any monic polynomial $h \in \Phi$ let $h(X) = X^n - c_1 \cdot X^{n-1} + \cdots + (-1)^n c_n$ where $c_i \in \mathbb{F}_q$ for all $i \in [n]$. Then we have $\lambda(h) = \psi(c_n) \cdot \chi(c_1)$. Clearly λ is a multiplicative map on G. Now for a given (c_1, c_k) pair there are q^{n-2} polynomials where same (c_1, c_n) occurs. Hence

$$\sum_{h \in \Phi_k} \lambda(h) = q^{n-2} \sum_{c_1 \in \mathbb{F}_q} \sum_{c_n \in \mathbb{F}_q^*} \psi(c_n) \cdot \chi(c_1) = q^{n-2}\left(\sum_{c_n \in \mathbb{F}_q}\right)\left(\sum_{c_1 \in \mathbb{F}_q} \chi(c_1)\right)$$

Since ψ and χ are non-trivial by Theorem 3.2.2 and Theorem 3.2.5 we have $\sum_{h \in \Phi_k} \lambda(h) = 0$ for $n > 1$.

Now $\Phi(0) = \{1\}$ and $\Phi(1) = \{x - \alpha \colon \alpha \in \mathbb{F}_q^*\}$. Hence

$$\sum_{h \in \Phi_1} \lambda(h) = \sum_{\alpha \in \mathbb{F}_q^*} \psi(\alpha) \cdot \chi(\alpha) = G(\psi, \chi)$$

Therefore

$$\overline{L}(z, \lambda) = 1 + G(\psi, \chi) \cdot z \qquad \text{and} \qquad L(s, \chi) = 1 + q^{-s} \cdot G(\psi, \chi)$$

Therefore root of $\overline{L}(z, \lambda)$ is $-G(\psi, \chi)$.

Now we will calculate L_r where $r = [E \colon \mathbb{F}_q]$. From the definition of L_r in previous section we have

$$L_r = \sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid r}} \deg(h) \cdot \chi(h)^{r/\deg(h)} = \sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid r}} \deg(h) \cdot \lambda\left(h^{r/\deg(h)}\right)$$

Now for any irreducible h let γ be a root of h in the splitting field of f. Let

$$h^{r/\deg(f)} = x^r - c_1 \cdot x^{r-1} + \cdots + (-1)^k c_r = \prod_{i=o}^{r-1}\left(X - \gamma^{q^i}\right)$$

Hence $c_1 = \mathrm{Tr}_{E/\mathbb{F}_q}(\gamma)$ and $c_r = \mathrm{N}_{E/\mathbb{F}_q}(\gamma)$. Hence we have

$$\lambda\left(h^{r/\deg(h)}\right) = \psi(c_r) \cdot \chi(c_1) = \psi\left(\mathrm{N}_{E/\mathbb{F}_q}(\gamma)\right) \cdot \chi\left(\mathrm{Tr}_{E/\mathbb{F}_q}(\gamma)\right) = \psi^{(r)}(\gamma) \cdot \chi^{(r)}(\gamma)$$

Now we can express $\deg(h) \cdot \lambda\left(h^{r/\deg(h)}\right) = \sum_{\gamma \in E, h(\gamma)=0} \psi^{(r)}(\gamma) \cdot \chi^{(r)}(\gamma)$. Therefore

$$L_r = \sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid k}} \deg(h) \cdot \lambda\left(h^{k/\deg(h)}\right) = \sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid k}} \sum_{\gamma \in E, h(\gamma)=0} \psi^{(r)}(\gamma) \cdot \chi^{(r)}(\gamma)$$

Now all irreducible polynomials of $\mathbb{F}_q$ whose degree divides r with non-zero constant their roots in E are disjoint sets and partition E^*. Therefore we have

$$L_r = \sum_{\gamma \in E^*} \psi^{(r)}(\gamma) \cdot \chi^{(r)}(\gamma) = G(\psi^{(r)}, \chi^{(r)})$$

Since by Theorem 3.5.10 we have $L_r = -(-G(\psi, \chi))^r$. So we have the theorem. ■

An analogous lifting phenomenon holds for Jacobi sums. Using the relation between Jacobi and Gaussian sums established in Theorem 3.4.3 together with the Davenport-Hasse Theorem, one can compare the Jacobi sum of a tuple of characters over $\mathbb{F}_q$ with the Jacobi sum of their lifts to a finite extension. The following theorem makes this precise.

Theorem 3.6.2

Let $\lambda_1, \ldots, \lambda_k \in \mathcal{M}_q$ be multiplicative characters of $\mathbb{F}_q$, not all of which are trivial. Suppose $\lambda_1, \ldots, \lambda_k$ are lifted to characters $\lambda_1(r), \ldots, \lambda_k^{(r)}$ respectively, of the finite extension field E of $\mathbb{F}_q$ with $[E : \mathbb{F}_q] = r$. Then

$$J\left(\lambda_1^{(r)}, \ldots, \lambda_k^{(r)}\right) = (-1)^{(r-1)(k-1)} J(\lambda_1, \ldots, \lambda_k)^r$$

Proof: Now if any λ_i is trivial then the lifted character $\lambda^{(r)}$ is also trivial over E. If Then by Theorem 3.4.6(ii) we get both sides to be 0. So suppose $\lambda_1, \ldots, \lambda_k$ are non-trivial. Now we will use Theorem 3.4.3. Let $\chi \in \mathcal{X}_q$ be any non-trivial additive character of $\mathbb{F}_q$. So let $\chi^{(r)}$ be the lifted character of χ to E. If $\prod_{i=1}^k \lambda_i$ is non-trivial then we have

$$\begin{aligned}
J\left(\lambda_1^{(r)}, \ldots, \lambda_k^{(r)}\right) &= \frac{\prod_{i=1}^k G\left(\lambda_i^{(r)}, \chi^{(r)}\right)}{G\left(\prod_{i=1}^k \lambda_i^{(r)}, \chi^{(r)}\right)} \\
&= \frac{\prod_{i=1}^k (-1)^{r-1} G(\lambda_i, \chi)^r}{(-1)^{r-1} G\left(\prod_{i=1}^k \lambda_i, \chi\right)^r} && \text{[By } \textit{Davenport} - \textit{Hasse Theorem}\text{]} \\
&= (-1)^{(s-1)(r-1)} \left[\frac{\prod_{i=1}^k G(\lambda_i, \chi)}{G\left(\prod_{i=1}^k \lambda_i, \chi\right)}\right]^r \\
&= (-1)^{(s-1)(r-1)} J(\lambda_1, \ldots, \lambda_k)^r
\end{aligned}$$

Now the only case remaining is when $\prod_{i=1}^k \lambda_i$ is trivial. In that case we have

$$\begin{aligned}
J\left(\lambda_1^{(r)}, \ldots, \lambda_k^{(r)}\right) &= -\frac{1}{q^r} \prod_{i=1}^k G\left(\lambda_i^{(r)}, \chi^{(r)}\right) \\
&= -\frac{1}{q^s} \cdot (-1)^{(r-1)\cdot k} \prod_{i=1}^k G(\lambda_i, \chi)^r && \text{[By } \textit{Davenport} - \textit{Hasse Theorem}\text{]} \\
&= (-1)^{(r-1)(k-1)} \cdot J(\lambda_1, \ldots, \lambda_k)
\end{aligned}$$

So we have the theorem. ■

3.7 Kloosterman Sums

We now turn to a genuinely different family of exponential sums, introduced by Hendrik Kloosterman in 1926 in his refinement of the Hardy–Littlewood circle method for the representation of integers by positive definite quaternary quadratic forms. Instead of twisting two characters of different types, a Kloosterman sum twists a single additive character χ with the *inversion* map $\gamma \mapsto \gamma^{-1}$ on $\mathbb{F}_q^*$, producing a sum of the form $\sum_\gamma \chi(a\gamma + b\gamma^{-1})$.

Definition 3.7.1: Kloosterman Sums

Let $\chi \in \mathcal{X}_q$ be a non-trivial additive character of $\mathbb{F}_q$ and let $a, b \in \mathbb{F}_q$. Then the Kloosterman Sum *associated to χ, a, b is defined as*

$$K(\chi; a, b) = \sum_{\gamma \in \mathbb{F}_q^*} \chi(a \cdot \gamma + b \cdot \gamma^{-1})$$

When one or both of the parameters a, b vanish, the Kloosterman sum collapses to an additive character sum on $\mathbb{F}_q^*$, which we have already computed in Theorem 3.2.2. If both $a = b = 0$ then for all $\gamma \in \mathbb{F}_q^*$, $\chi(a \cdot \gamma + b \cdot \gamma^{-1}) = \chi(0) = 1$. Hence $K(\chi; 0, 0) = q - 1$.

If exactly one of a, b is zero then we have either

$$K(\chi; a, b) = \sum_{\gamma \in \mathbb{F}_q^*} \chi(a \cdot \gamma) = -1 \text{ or } K(\chi; a, b) = \sum_{\gamma \in \mathbb{F}_q^*} \chi(b \cdot \gamma^{-1}) = -1$$

So in both cases $K(\chi; a, b) = -1$. So we have the following observations:

Observation 3.14. *Let $\chi \in \boldsymbol{\chi}_q$ be a non-trivial additive character of $\mathbb{F}_q$ and $a, b \in \mathbb{F}_q$ such that $ab = 0$. Then*

(i) $K(\chi; 0, 0) = q - 1$ if both $a = b = 0$;

(ii) if exactly one of a, b is 0 then $K(\chi; a, b) = -1$;

Now another property to notice is Kloosterman sum is always real since

$$\overline{K(\chi; a, b)} = \sum_{\gamma \in \mathbb{F}_q^*} \overline{\chi}(a \cdot \gamma + b \cdot \gamma^{-1}) = \sum_{\gamma \in \mathbb{F}_q^*} \chi(-(a \cdot \gamma + b \cdot \gamma^{-1})) = \sum_{\gamma \in \mathbb{F}_q^*} \chi(a \cdot (-\gamma) + b \cdot (-\gamma)^{-1}) = K(\chi; a, b)$$

Therefore we have another observation:

Observation 3.15. *For all additive characters $\chi \in \boldsymbol{\chi}_q$ and $a, b \in \mathbb{F}_p$, the Kloosterman Sum is real i.e. $\overline{K(\chi; a, b)} = K(\overline{\chi}; a, b) = K(\chi; -a, -b) = K(\chi; a, b)$.*

Now we can express Kloosterman sums as product of two additive characters like Jacobi sums is for product of multiplicative characters. For any $\chi \in \boldsymbol{\chi}_q$ let $\chi = \chi_c$ for some $c \in \mathbb{F}_q$ as defined in Theorem 3.2.1. Then

$$K\left(\chi_c; a, b\right) = \sum_{\gamma \in \mathbb{F}_q^*} \chi_c\left(a \cdot \gamma + b \cdot \gamma^{-1}\right) = \sum_{\gamma \in \mathbb{F}_q^*} \chi_{a \cdot c}(\gamma) \cdot \chi_{-b \cdot c}(\gamma) =: K\left(\chi_{ac}, \overline{\chi}_{bc}\right)$$

Observation 3.16. *The Kloosterman sum can equivalently be viewed as two-additive-character sum: for any $a, b, c \in \mathbb{F}_q$ and $\chi_a, \chi_b, \chi_c \in \boldsymbol{\chi}_q$ (in the notation of Theorem 3.2.1),*

$$K\left(\chi_a, \chi_b\right) := K\left(\chi_1; a, -b\right) \qquad \text{and} \qquad K\left(\chi_c; a, -b\right) = K\left(\chi_{ac}, \chi_{bc}\right)$$

When the ground field has odd characteristic there is a useful alternative expression for $K(\chi; a, b)$ as a sum of a single additive character against the quadratic character η evaluated on a quadratic polynomial. This representation is often the starting point for asymptotic bounds on Kloosterman sums in arithmetic contexts.

Lemma 3.7.1

Let $\chi \in \boldsymbol{\chi}_q$ be a non-trivial additive character of $\mathbb{F}_q$ with q odd, let η be the quadratic character of $\mathbb{F}_q$ (extended by $\eta(0) = 0$), and let $a, b \in \mathbb{F}_q$ be not both zero. Then

$$K(\chi; a, b) = \sum_{d \in \mathbb{F}_q} \chi(d) \cdot \eta(d^2 - 4ab)$$

Proof: If one of a, b is zero then by Observation 3.14(ii) we have $K(\chi; a, b) = -1$. And then at the right hand side we have the sum over $\eta(\alpha^2 - 4ab) = \eta(\alpha^2) = 1$ and therefore the sum is of $\chi(\alpha)$ over all non-zero alpha which by Theorem 3.2.2(i) gives $q - 1 = -1$. Therefore in this case the lemma is satisfied. So now assume $a \cdot b \neq 0$.

Let for any $\gamma \in \mathbb{F}_q^*$, denote $d = a \cdot \gamma + b \cdot \gamma^{-1}$. Now for any $d \in \mathbb{F}_q$ let $Z(d)$ is the number of $\gamma \in \mathbb{F}_q^*$ such that $d = a \cdot \gamma + b \cdot \gamma^{-1}$. Now for any $d \in \mathbb{F}_q$,

$$a \cdot \gamma + b \cdot \gamma^{-1} = d \iff a \cdot \gamma^2 - d \cdot \gamma + b = 0$$

Therefore if $P_d(X) = a \cdot X^2 - d\dot{X} + b \in \mathbb{F}_q[X]$ be a polynomial then $Z(d)$ = number of zeros of $P_d(X)$. Hence $Z(d) \leq 2$. Let Δ_d is the discriminant of P_d i.e. $\Delta_d = d^2 - 4ab$. Hence If $\Delta_d = 0$ then $Z(d) = 1$. If a non-zero square root of Δ_d exists in $\mathbb{F}_q$ then Δ_d is a quadratic residue i.e. $\eta(\Delta_d) = 1$ then $Z(d) = 2$ and if it doesn't exists then Δ_d is a quadratic non-residue i.e. $\eta(\Delta_d) = -1$ then $Z(d) = 0$. Therefore together we can write $Z(d) = \eta(\Delta_d) + 1$ with the extended definition of η by setting $\eta(0) = 0$. Therefore we have

$$K(\chi; a, b) = \sum_{d \in \mathbb{F}_q} \chi(d) \cdot Z(d) = \sum_{d \in \mathbb{F}_q} \chi(d) \cdot \cdot \eta(\Delta_d) = \sum_{d \in \mathbb{F}_q} \chi(d) \cdot \cdot \eta(d^2 - 4ab)$$

Hence we have the lemma. ■

One of the central structural result for Kloosterman sums asserts that the sequence of lifted sums $K(\chi^{(s)}; a, b)$, viewed as a function of s, is controlled by just two complex numbers ω_1, ω_2 – exactly the reciprocal roots of the associated L-function. This is the Kloosterman analogue of the Davenport-Hasse Theorem. We will use again the *Dirichlet L-functions* in section 3.5 to prove the following theorem

Theorem 3.7.2

Let $\chi \in \boldsymbol{\chi}_q$ be a non-trivial additive character of $\mathbb{F}_q$ and let $a, b \in \mathbb{F}_q$ with $ab \neq 0$. Suppose χ is lifted to character $\chi^{(r)}$ of the finite extension field E of $\mathbb{F}_q$ with $[E\colon \mathbb{F}_q] = r$. Then there exist complex numbers w_1, w_2 (depending only on χ, a, b), which are either complex conjugates of each other or both real, such that for every positive integer r

$$K(\chi^{(r)}; a, b) = \sum_{\gamma \in \mathbb{F}_{q^r}^*} \chi^{(r)}(a\gamma + b\gamma^{-1}) = -w_1^r - w_2^r,$$

Proof: Let G be the set of all rational functions $r(X) \in \mathbb{F}_q(X)$ such that for any $r \in G$, $r(X) = {}^{h_1(X)}/_{h_2(X)}$ where $h_1, h_2 \in \mathbb{F}_q[x]$ and both h_1, h_2 are monic. Let $\overline{G}$ the subgroup of G such that for any $r \in \overline{G}$, $r(X) = {}^{h_1(X)}/_{h_2(X)}$ where h_1, h_2 have non-zero constant term. We will define a multiplicative function $\lambda : G \to \mathbb{C}$ such that $|\lambda(r)| \leq 1$ for all $r \in G$ and $\lambda(1) = 1$. Let $r \in \overline{G}$ where $r(X) = \frac{h_1(X)}{h_2(X)}$ and in common splitting field E of h_1, h_2 they factor as

$$h_1(X) = \prod_{i=1}^{d_1}(X - \alpha_i) \qquad \text{and} \qquad h_2(X) = \prod_{j=1}^{d_2}(X - \beta_j)$$

Then we will define λ using χ in the following way:

$$\lambda(r) = \chi\left[a\left(\sum_{i=1}^{d_1}\alpha_i - \sum_{j=1}^{d_2}\beta_j\right) + b\left(\sum_{i=1}^{d_1}\frac{1}{\alpha_i} - \sum_{j=1}^{d_2}\frac{1}{\beta_j}\right)\right] \tag{3.10}$$

Then if $h \in \Phi_n$ such that $h(X) = X^n - c_1 \cdot X^{n-1} + \cdots + (-1)^n \cdot c_n$ then $\lambda(h) = \chi\left(a \cdot c_1 + b \cdot c_{k-1} \cdot c_k^{-1}\right)$. Now it is enough to show that the corresponding coefficients are additive if two polynomials are multiplied since χ is an additive character. Let $h_1 \in \Phi_{n_1}$ and $g_2 \in \Phi_{n_2}$ and

$$g_1(X) = X^{n_1} + \sum_{i=1}^{n_1}(-1)^i \cdot c_{1,i} \cdot X^{n_1 - i}, \quad g_2(X) = X^{n_2} + \sum_{j=2}^{n_2}(-2)^j \cdot c_{2,j} \cdot X^{n_2 - j} \text{ and } g_1 \cdot g_2 = X^{n_1+n_2} + \sum_{t=1}^{n_1+n_2}(-1)^t \cdot c_j \cdot X^{n_1+n_2-t}$$

Then by comparing coeffieicnets we have

$$c_1 = c_{1,1} + c_{2,1}, \quad c_{n_1+n_2} = c_{1,n_1} \cdot c_{2,n_2}, \quad \text{and} \quad c_{n_1+n_2-1} = c_{1,n_1} \cdot c_{2,n_2-1} + c_{1,n_1-1} \cdot c_{2,n_2}$$

Hence λ is indeed a multiplicative function. Now we extend λ to G by setting $\lambda(r) = 0$ for all $h \notin \overline{G}$.

Now for any $k \in \mathbb{Z}_0$,if (c_1, c_{k-1}, c_k) is fixed where $c_k \neq 0$, there are q^{k-3} possible choices for $c_2, \ldots, c_{k-2}$ such that all polynomials with coefficients c_i's give same value for $\chi(a \cdot c_1 + b \cdot c_{k-1} \cdot c_k^{-1}$. So we will show that for $k \geq 3$, $\sum_{h \in \Phi_k} \lambda(h) = 0$. So

$$\begin{aligned}
\sum_{g \in \Phi_k} \lambda(g) &= \sum_{c_1, \ldots, c_{k-1}} \sum_{c_k \in \mathbb{F}_q^*} \chi(a \cdot c_1 + b \cdot c_{k-1} \cdot c_k^{-1}) \\
&= q^{k-3} \sum_{c_1, c_{k-1} \in \mathbb{F}_q} \sum_{c_k \in \mathbb{F}_q^*} \chi(a \cdot c_1 + b \cdot c_{k-1} \cdot c_k^{-1}) \\
&= q^{k-3} \left(\sum_{c_1 \in \mathbb{F}_q} \chi(a \cdot c_1) \right) \left(\sum_{c_{k-1} \in \mathbb{F}_q} \sum_{c_k \in \mathbb{F}_q^*} \chi(b \cdot c_{k-1} \cdot c_k^{-1}) \right) = 0 \qquad \text{[By } \textit{Theorem } 3.2.2(i)\text{]}
\end{aligned}$$

Hence we have $\overline{L}(z, \lambda) = 1 + z \sum_{h \in \Phi_1} \lambda(h) + z^2 \sum_{h \in \Phi_2} \lambda(h)$. Now $\Phi_1 = \{X - \gamma \mid \gamma \in \mathbb{F}_q^*\}$. Therefore we have

$$\sum_{h \in \Phi_1} \lambda(h) = \sum_{\gamma \in \mathbb{F}_q^*} \chi(a \cdot \gamma + b \cdot \gamma^{-1}) = K(\chi; a, b)$$

And furthermore for Φ_2 we get

$$\begin{aligned}
\sum_{h \in \Phi_2} \lambda(h) &= \sum_{c_1 \in \mathbb{F}_q} \sum_{c_2 \in \mathbb{F}_q^*} \lambda(X^2 - c_1 \cdot X + c_2) \\
&= \sum_{c_1 \in \mathbb{F}_q} \sum_{c_2 \in \mathbb{F}_q^*} \chi(a \cdot c_1 + b \cdot c_1 \cdot c_2^{-1}) \\
&= \sum_{c_1 \in \mathbb{F}_q} \sum_{c_2 \in \mathbb{F}_q^*} \chi(c_1(a + b \cdot c_2^{-1}))
\end{aligned}$$

So if $c_2 = -a^{-1}b$ then $a + b \cdot c_2^{-1} = 0$ and hence $\sum_{c_1 \in \mathbb{F}_q} \chi(0) = q$ and if $c_2 \neq -a^{-1}b$ then $\sum_{c_1 \in \mathbb{F}_q} \chi(c_1(a + b \cdot c_2^{-1})) = 0$ by Theorem 3.2.2(i). So therefore $\sum_{h \in \Phi_2} \lambda(h) = q$ and thus we get $\overline{L}(z, \lambda) = 1 + k(\chi; a, b) \cdot z + q \cdot z^2$. Let $w_1, w_2 \in \mathbb{C}$ which are either complex conjugates or both real such that

$$\overline{L}(z, \lambda) = (1 - w_1 \cdot z)(1 - w_2 \cdot z)$$

Now we will calculate L_r where $r = [E \colon \mathbb{F}_q]$. From the definition of L_r in previous section we have

$$L_r = \sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid r}} \deg(h) \cdot \chi(h)^{r/\deg(h)} = \sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid r}} \deg(h) \cdot \lambda\left(h^{r/\deg(h)}\right)$$

Now for any irreducible h let γ be a root of h in the splitting field of f. Let

$$h^{r/\deg(f)} = x^r - c_1 \cdot x^{r-1} + \cdots + (-1)^k c_r = \prod_{i=o}^{r-1} \left(X - \gamma^{q^i}\right)$$

Hence $c_1 = \mathrm{Tr}_{E/\mathbb{F}_q}(\gamma)$ and $c_r = \mathrm{N}_{E/\mathbb{F}_q}(\gamma)$. Hence we have

$$\lambda\left(h^{r/\deg(h)}\right) = \psi(c_r) \cdot \chi(c_1) = \psi\left(\mathrm{N}_{E/\mathbb{F}_q}(\gamma)\right) \cdot \chi\left(\mathrm{Tr}_{E/\mathbb{F}_q}(\gamma)\right) = \psi^{(r)}(\gamma) \cdot \chi^{(r)}(\gamma)$$

Now we can express $\deg(h) \cdot \lambda\left(h^{r/\deg(h)}\right) = \sum_{\gamma \in E, h(\gamma)=0} \psi^{(r)}(\gamma) \cdot \chi^{(r)}(\gamma)$. Therefore

$$L_r = \sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid k}} \deg(h) \cdot \lambda\left(h^{k/\deg(h)}\right) = \sum_{\substack{h \in \mathrm{Irr}(\Phi), \\ \deg(h) \mid k}} \sum_{\gamma \in E, h(\gamma)=0} \psi^{(r)}(\gamma) \cdot \chi^{(r)}(\gamma)$$

Now all irreducible polynomials of $\mathbb{F}_q$ whose degree divides r with non-zero constant their roots in E are disjoint sets and partition E^*. Therefore we have

$$L_r = \sum_{\gamma \in E^*} \psi^{(r)}(\gamma) \cdot \chi^{(r)}(\gamma) = G(\psi^{(r)}, \chi^{(r)})$$

Since by Theorem 3.5.10 we have $L_r = -w_1^r - w_2^r$. So we have the theorem. ■

Assuming w_1, w_2 as above, every lifted Kloosterman sum is a symmetric function of w_1, w_2, which by Waring's identities can be rewritten purely in terms of $K = K(\chi; a, b)$ and q. This gives the following closed-form reduction of $K(\chi^{(s)}; a, b)$ to the ground-field sum.

Lemma 3.7.3

Let $\chi \in \mathcal{X}_q$ be a non-trivial additive character of $\mathbb{F}_q$ and let $a, b \in \mathbb{F}_q$ with $ab \neq 0$. Then for every positive integer s,

$$K\left(\chi^{(s)}; a, b\right) = \sum_{j=0}^{\lfloor s/2 \rfloor} (-1)^{s-j-1} \cdot \frac{s}{s-j} \cdot \binom{s-j}{j} \cdot q^j \cdot K(\chi; a, b)^{s-2j}.$$

Proof: Consider the polynomial $X_1^r + X_2^r$. Now by Waring's Formula we can express this polynomial in terms of elementary symmetric polynomials $\mathrm{ESym}_1, \ldots, \mathrm{ESym}_r$ which gives

$$\begin{aligned} X_1^r + X_2^r &= \sum_{\substack{d_1, d_2 \in \mathbb{Z}_0 \\ d_1 + 2d_2 = r}} (-1)^{d_2} \frac{r(d_1 + d_2 - 1)!}{d_1! d_2!} \mathrm{ESym}_1(X_1, X_2)^{d_1} \cdot \mathrm{ESym}_2(X_1, X_2)^{d_2} \\ &= \sum_{j=0}^{\lfloor r/2 \rfloor} (-1)^j \frac{(r-j-1)! r}{j!(s-2j)!} \mathrm{ESym}_1(X_1, X_2)^{s-2j} \cdot \mathrm{ESym}_2(X_1, X_2)^j \\ &= \sum_{j=0}^{\lfloor r/2 \rfloor} (-1)^j \binom{r-j}{j} \frac{r}{s-j} \mathrm{ESym}_1(X_1, X_2)^{s-2j} \cdot \mathrm{ESym}_2(X_1, X_2)^j \end{aligned}$$

Now we take $X_1 = w_1$ and $X_2 = w_2$ obtained from Theorem 3.7.2 we get $w_1^k + w_2^k = -K(\chi^{(k)}; a, b)$ for any $k \in \mathbb{N}$ and $w_1 \cdot w_2 = q$. So by this replacing in the equation we get the result. ■

This lemma gives a relation between powers of Kloosterman sum and lifted Kloosterman sums. We can obtain a recursive relation between different lifts of Kloosterman sums very easily using this identiy

$$w_1^r + w_2^r = (w_1^{r-1} + w_2^{r-1})(w_1 + w_2) - (w_1^{r-2} + w_2^{r-2}) w_1 w_2$$

which gives us

$$K(\chi^{(r)}; a, b) = -K(\chi^{(r-1)}; a, b) \cdot K(\chi; a, b) - q \cdot K(\chi^{(r-2)}; a, b)$$

and so we have the corollary.

Corollary 3.7.4

Let $\chi \in \mathcal{X}_q$ be a non-trivial additive character of $\mathbb{F}_q$ and let $a, b \in \mathbb{F}_q$ with $ab \neq 0$. Then for every positive integer s, the lifted Kloosterman sums satisfy the recurrence

$$K\left(\chi^{(s)}; a, b\right) = -K\left(\chi^{(s-1)}; a, b\right) \cdot K(\chi; a, b) - K\left(\chi^{(s-2)}; a, b\right) \cdot q \qquad \text{for all } s \geq 2,$$

with the convention $K(\chi^{(0)}; a, b) := -2$ (and noting $K\left(\chi^{(1)}; a, b\right) = K(\chi; a, b)$).

3.8 Character Sums via the Quadratic Character

For certain special characters, the associated Gaussian sums can be evaluated explicitly. A very special multiplicative character to study is the quadratic character, η. We get a very celebrated formula for the Gaussian sum of η and the canonical additive character.

Theorem 3.8.1

Let $\mathbb{F}_q$ be a finite field with $q = p^r$ where p is an odd prime and $r \in \mathbb{N}$. Let η be the quadratic character of $\mathbb{F}_q$ and let χ_1 be the canonical additive character of $\mathbb{F}_q$. Then

$$G(\eta, \chi_1) = \begin{cases} (-1)^{r-1} \cdot q^{1/2} & \text{if } p \equiv 1 \bmod 4 \\ (-1)^{r-1} \cdot i^r \cdot q^{1/2} & \text{if } p \equiv 3 \bmod 4 \end{cases}$$

In other words,

$$G(\eta, \chi_1) = (-1)^{r-1} \cdot i^{\frac{(p-1)^2}{4} r} \cdot q^{1/2}$$

Proof: Since η always takes values in ± 1 we have $\eta = \overline{\eta}$. Hence by using Lemma 3.3.2(iv) we have $G(\eta, \chi_1)^2 = \eta(-1)q$. Now if $q \equiv 1 \bmod 4$ then $\eta(-1) = 1$ and if $q \equiv 3 \bmod 4$ then $\eta(-1) = -1$. Hence

$$G(\eta, \chi_1) = \begin{cases} \pm\, q^{1/2} & \text{if } q \equiv 1 \bmod 4 \\ \pm\, i \cdot q^{1/2} & \text{if } q \equiv 3 \bmod 4 \end{cases} \tag{3.11}$$

Since $i^{(p-1)^2/4} = \begin{cases} 1 & \text{if } q \equiv 1 \bmod 4 \\ 0 & \text{if } q \equiv 3 \bmod 4 \end{cases}$

Observation 3.17. *For any finite field $\mathbb{F}_q$, $G(\eta, \chi_1) = \pm\, i^{(q-1)^2/4} \cdot q^{1/2}$*

Now comes the hardest part of the proof that is getting the signs correctly. We will first prove this for $r = 1$ case then we will go for $r > 1$. Now we have the observation

Let W be the vector space of all complex valued functions on $\mathbb{F}_p^*$. It is a $(p-1)$-dimensional vector space over $\mathbb{C}$. So $\mathcal{B} := \{f_1, \ldots, f_{p-1}\}$ where $f_j(k) = 1$ if $j = k$ and other wise 0 forms a basis of W for all $j \in [p-1]$. Now by using the orthogonal properties of multiplicative characters in Theorem 3.2.5 the set $\mathcal{B}_{\mathcal{M}_q} := \{\psi_0, \ldots, \psi_{p-2}\}$ as defined in Lemma 3.2.3, also forms a basis of W.

Let ζ_p is the pth root of unity i.e. $\zeta_p = e^{2\pi i/p}$. Hence if $\chi_k \in \boldsymbol{\chi}_p$ be an additive character of $\mathbb{F}_p$ as defined in Theorem 3.2.1, then for all $j \in \mathbb{F}_p$, $\chi_k(j) = \zeta_p^{jk}$. Then define the linear operator $T : W \to W$ in the following way:

$$\forall\, h \in W,\ \forall\, k \in [p-1] \qquad (Th)(k) = \sum_{j=1}^{p-1} \zeta_p^{jk} h(j) \tag{3.12}$$

Therefore for all basis elements ψ_i

$$(T\psi_i)(k) = \sum_{j=1}^{p-1} \zeta_p^{jk} \cdot \psi_i(k) = \sum_{j=1}^{p-1} \chi_k(j) \cdot \psi_i(k) = G(\psi, \chi_k) = \overline{\psi_i}(k) \cdot G(\psi_i, \chi_1) \quad \forall\, k \in \mathbb{F}_p^*$$

where the last equality comes from Lemma 3.3.2(i). Hence $T\psi_i = G(\psi_i, \chi_1) \cdot \overline{\psi_i}$ for every multiplicative characters of $\mathbb{F}_p$.

Since $\overline{\psi} = \psi$ only for the trivial character and quadratic character, the matrix for T in the basis $\mathcal{B}_{\mathcal{M}_q} = \{\psi_0, \ldots, \psi_{p-2}\}$ contains two diagonal entries, $G(\psi_0, \chi_1) = -1$ and $G(\eta, \chi_1)$ for ψ_0 and $\eta = \psi_{p-1/2}$. For all other characters we can divide them into groups of pairs where the characters in each pair are conjugates of each other. Then for each pair $(\psi, \overline{\psi})$ we have the following block matrix in the matrix of T.

$$\begin{bmatrix} 0 & G(\overline{\psi}, \chi_1) \\ G(\psi, \chi_1) & 0 \end{bmatrix}$$

Therefore in the determinant of the matrix T each block contributes $-G(\psi, \chi_1)G(\overline{\psi}, \chi_1) = -\psi(-1)p$ by Lemma 3.3.2(iv). So we get

$$\det(T) = (-1) \cdot G(\eta, \chi_1) \cdot (-p)^{(p-3)/2} \prod_{j=1}^{\frac{p-3}{2}} \psi_j(-1)$$

Let $\psi \in \boldsymbol{\mathcal{M}}_p$ is the principle multiplicative character of $\mathbb{F}_p$. Then $\psi_j = \psi^j$. Therefore $\psi_j(-1) = \psi^j(-1) = (-1)^j$ as $\psi(-1) = -1$. Hence we have

$$\begin{aligned}
\det(T) &= -G(\eta, \chi_1) \cdot (-p)^{(p-3)/2} \prod_{j=1}^{\frac{p-3}{2}} (-1)^j \\
&= -G(\eta, \chi_1) \cdot (-p)^{(p-3)/2} \cdot (-1)^{\sum_{j=1}^{(p-3)/2} j} \\
&= (-1)^{(p-1)/2} \cdot G(\eta, \chi_1) \cdot p^{(p-3)/2} \cdot (-1)^{(p-1)(p-3)/8} \\
&= (-1)^{(p-1)/2} \cdot \left(\pm\, i^{(p-1)^2)/4} \cdot p^{1/2}\right) \cdot p^{(p-3)/2} \cdot (-1)^{(p-1)(p-3)/8} && \text{[By \textit{Observation} 3.17]} \\
&= \pm(-1)^{(p-1)/2} \cdot i^{\frac{(p-1)^2)}{4} + \frac{(p-1)(p-3)}{4}} \cdot p^{(p-2)/2} \\
&= \pm(-1)^{(p-1)/2} \cdot i^{(p-1)(p-2)/2} \cdot p^{(p-2)/2}
\end{aligned}$$

So we get

$$\det(T) = \pm(-1)^{(p-1)/2} \cdot i^{(p-1)(p-2)/2} \cdot p^{(p-2)/2} \tag{3.13}$$

Now we will compute $\det(T)$ utilizing the basis $\mathcal{B} = \{f_1, \ldots, f_{p-1}\}$. From the construction of T in (3.12) we get the $(j,k)^{th}$ element of the matrix of T in the $\{f_1, \ldots, f_{p-1}\}$ basis to be ζ_p^{jk} for any $0 \le j, k \le p-1$. So we have

$$\begin{aligned}
\det(T) &= \det\left(\left\{\zeta_p^{jk}\right\}_{1 \le j,k \le p-1}\right) \\
&= \det\left(\left\{\zeta_p^{j}\zeta_p^{j(k-1)}\right\}_{1 \le j,k \le p-1}\right) \\
&= \zeta_p^{1+2+\cdots+(p-1)} \det\left(\left\{\zeta_p^{j(k-1)}\right\}_{1 \le j,k \le p-1}\right) \\
&= \det\left(\left\{\zeta_p^{j(k-1)}\right\}_{1 \le j,k \le p-1}\right) && \left[\text{As } p \mid \frac{p(p-1)}{2} = 1 + 2 + \cdots + (p-1)\right]
\end{aligned}$$

Now the matrix $\left\{\zeta_p^{j(k-1)}\right\}_{1 \le j,k \le p-1}$ is the *Vandermonde* matrix $V(\zeta_p, \zeta_p^2, \ldots, \zeta_p^{p-1})$. Therefore

$$\det(T) = \prod_{1 \le j < k \le p-1} \left(\zeta_p^k - \zeta_p^j\right)$$

We will show that

$$(-1)^{(p-1)/2} \cdot i^{(p-1)(p-2)/2} \cdot p^{(p-2)/2} = \prod_{1 \le j < k \le p-1} \left(\zeta_p^k - \zeta_p^j\right) \tag{3.14}$$

So let $\zeta_{2p} = e^{\pi i/p}$. Hence $\zeta_{2p}^2 = \zeta_p$. By the properties of complex numbers we know for all $t \in \mathbb{Z}$, $\zeta_{2p}^t = \cos\left(\frac{t\pi}{p}\right) + i \sin\left(\frac{t\pi}{p}\right)$.

Hence $\zeta_{2p}^{t} - \zeta_{2p}^{-t} = 2i \sin\left(\frac{t\pi}{p}\right)$. So we get

$$\begin{aligned}
\det(T) &= \prod_{1\le j<k\le p-1} \left(\zeta_p^k - \zeta_p^j\right) \\
&= \prod_{1\le j<k\le p-1} \left(\zeta_{2p}^{2k} - \zeta_{2p}^{2j}\right) \\
&= \prod_{1\le j<k\le p-1} \zeta_{2p}^{j+k}\left(\zeta_{2p}^{k-j} - \zeta_{2p}^{-(k-j)}\right) \\
&= \left[\prod_{1\le j<k\le p-1} \zeta_{2p}^{j+k}\right]\left[\prod_{1\le j<k\le p-1}\left(2i\sin\frac{\pi(k-j)}{p}\right)\right] \\
&= \zeta_{2p}^{\sum_{1\le j<k\le p-1}(j+k)} \cdot i^{\binom{p-1}{2}} \prod_{1\le j<k\le p-1}\left(2\sin\frac{\pi(k-j)}{p}\right) \\
&= \zeta_{2p}^{\sum_{1\le j<k\le p-1}(j+k)} \cdot i^{(p-1)(p-2)/2} \prod_{1\le j<k\le p-1}\left(2\sin\frac{\pi(k-j)}{p}\right)
\end{aligned}$$

Now we calculate the power of ζ_{2p} in the first term.

$$\sum_{1\le j<k\le p-1}(j+k) = \sum_{k=2}^{p-1}\sum_{j=1}^{k-1}(j+k) = \sum_{k=2}^{p-1}\left(\frac{k(k-1)}{2} + k(k-1)\right) = \frac{3}{2}\sum_{k=2}^{p-1}(k^2+k) = \frac{p(p-1)(p-2)}{2}$$

Hence

$$\zeta_{2p}^{\sum_{1\le j<k\le p-1}(j+k)} = \left(\zeta_{2p}^{p}\right)^{(p-1)(p-2)/2} = (-1)^{(p-1)(p-2)/2} = \left((-1)^{p-2}\right)^{(p-1)/2} = (-1)^{(p-1)/2}$$

So we have

$$\det(T) = (-1)^{(p-1)/2} \cdot i^{(p-1)(p-2)/2} \cdot \prod_{1\le j<k\le p-1}\left(2\sin\frac{\pi(k-j)}{p}\right) \tag{3.15}$$

Notice if we compare this with (3.13), the first two terms are already same. If we can show that $\prod_{1\le j<k\le p-1} 2\sin\frac{\pi(k-j)}{p} > 0$ then we have (3.14).

Now since $j,k \in [p-1]$ and $k > j$ we have $k-j \in [p-1]$. Hence $\sin\frac{\pi(k-j)}{p} > 0$ for all $1 \le j < k \le p-1$. Hence

$$\prod_{1\le j<k\le p-1}\left(2\sin\frac{\pi(k-j)}{p}\right) > 0$$

So if we compare (3.15) with (3.13) and match the signs we get

$$\det(T) = (-1)^{(p-1)/2} \cdot i^{(p-1)(p-2)/2} \cdot p^{(p-2)/2}$$

So from this we get $G(\eta, \chi_1) = i^{(p-1)^2/4} \cdot p^{1/2}$ since at no point of our calculation the sign was changed.

So now we have established the theorem for $r = 1$. Now for $r > 1$ following subsection 3.2.3 we can lift the quadratic character of $\mathbb{F}_p$, η to a multiplicative character $\eta^{(r)}$ of $\mathbb{F}_q$ where $q = p^r$ and $\eta^{(r)} = \eta \circ N$ where N is the absolute norm function from $\mathbb{F}_q$ to $\mathbb{F}_p$. Since η is a quadratic character $\eta^{(r)}$ will be the quadratic character of $\mathbb{F}_q$. Similarly we can lift the canonical additive character χ_1 of $\mathbb{F}_p$ to the canonical additive character $\chi_q^{(r)}$ of $\mathbb{F}_q$ by $\chi_1^{(r)} = \chi_1 \circ \mathrm{Tr}$ where Tr is the absolute trace function from $\mathbb{F}_q$ to $\mathbb{F}_p$. Then by using Davenport-Hasse Theorem we get

$$G(\eta^{(r)}, \chi_1^{(r)}) = (-1)^{r-1} G(\eta, \chi_1)^r = (-1)^{r-1}\left[i^{(p-1)^2/4} \cdot p^{1/2}\right] = (-1)^r \cdot i^{\frac{(p-1)^2}{4}r} \cdot p^{r/2} = (-1)^r \cdot i^{\frac{(p-1)^2}{4}r} \cdot q^{1/2}$$

So we have the theorem. ■

The proof of the above theorem gives us another important result.

Lemma 3.8.2

If p is a odd prime and ζ_p is the pth root of unity. Let $V(\zeta_p, \zeta_p^2, \ldots, \zeta_p^{p-1})$ is the $p \times p$ Vandermonde Matrix. Then

$$\det\left(V(\zeta_p, \zeta_p^2, \ldots, \zeta_p^{p-1})\right) = (-1)^{(p-1)/2} \cdot i^{(p-1)(p-2)/2} \cdot p^{(p-2)/2}$$

As a striking application of the machinery developed around the quadratic character, we can now give a proof of one of the classical result of number theory – *Law of Quadratic Reciprocity.*

Theorem 3.8.3 Law of Quadratic Reciprocity

For any two distinct odd primes p, q we have

$$\left(\frac{p}{q}\right) \cdot \left(\frac{q}{p}\right) = (-1)^{(p-1)(q-1)/4}$$

Proof: Let χ_1 be the canonical additive character of $\mathbb{F}_p$ and η is the quadratic character of $\mathbb{F}_p$. Then by Lemma 3.3.2(iv) we gave $G^2(\eta, \chi_1) = (-1)^{(p-1)/2} \cdot p$. So

$$G^q(\eta, \chi_1) = G(\eta, \chi_1) \cdot \left(G^2(\eta, \chi_1)\right)^{(q-1)/2} = (-1)^{(p-1)(q-1)/4} \cdot p^{(q-1)/2} \cdot G(\eta, \chi_1) \tag{3.16}$$

Now we calculate $G^q(\eta, \chi_1)$ in different way.

$$\begin{aligned}
G^q(\eta, \chi_1) &= \left(\sum_{\alpha \in \mathbb{F}_p^*} \eta(\alpha) \cdot \chi_1(\alpha)\right)^q \\
&\equiv \sum_{\alpha \in \mathbb{F}_p^*} \eta^q(\alpha) \cdot \chi_q^q(\alpha) \bmod q \\
&\equiv \sum_{\alpha \in \mathbb{F}_p^*} \eta(\alpha) \cdot \chi_q(\alpha) \bmod q \\
&\equiv G(\eta, \chi_q) \bmod q \\
&\equiv \eta(q) G(\eta, \chi_1) \bmod q && \text{[By } \textit{Lemma 3.3.2(i)}\text{]}
\end{aligned} \tag{3.17}$$

Now we already obtained $\left(\frac{q}{p}\right)$ in the right hand side above and $(-1)^{(p-1)(q-1)/4}$ in (3.16). Since η and χ_1 are both non-trivial characters of $\mathbb{F}_p$ we have $|G(\eta, \chi_1)|^{=} p^{1/2}$ by Theorem 3.3.1 and so $G(\eta, \chi_1) \neq 0$. So now comparing (3.17) with (3.16) we get

$$(-1)^{(p-1)(q-1)/4} \cdot p^{(q-1)/2} \cdot G(\eta, \chi_1) \equiv \eta(q) \cdot G(\eta, \chi_1) \bmod q \iff (-1)^{(p-1)(q-1)/4} \cdot p^{(q-1)/2} \equiv \eta(q) \bmod q$$

By Fermat's Little Theorem we have $p^{q-1} \equiv 1 \bmod q$ as p and q are two distinct primes. So we have

$$\begin{aligned}
& (-1)^{(p-1)(q-1)/4} \cdot p^{(q-1)/2} \equiv \eta(q) \bmod q \\
\implies & (-1)^{(p-1)(q-1)/4} \cdot p^{(q-1)/2} \cdot p^{(q-1)/2} \equiv \eta(q) \cdot p^{(q-1)/2} \bmod q \\
\implies & (-1)^{(p-1)(q-1)/4} \equiv \eta(q) \cdot p^{(q-1)/2} \bmod q \\
\implies & (-1)^{(p-1)(q-1)/4} \equiv \left(\frac{q}{p}\right) \cdot \left(\frac{p}{q}\right) \bmod q
\end{aligned}$$

Since both sides are ± 1 we can remove the modulo q. And therefore we have the theorem. ■

We close the section with an elementary but remarkably clean evaluation of the quadratic character summed against a quadratic polynomial. Its value depends only on the leading coefficient and the discriminant, and it will play a basic role in many later arguments – most notably in extracting the quadratic character of the discriminant when counting solutions of quadratic equations over $\mathbb{F}_q$, and in the reformulation of Kloosterman sums in **??**.

Theorem 3.8.4

Let $f(X) = a_2X^2 + a_1X + a_0 \in \mathbb{F}_q[X]$ with q odd and $a_2 \neq 0$, let $\Delta = a_1^2 - 4a_0a_2$ denote the discriminant of f, and let η be the quadratic character of $\mathbb{F}_q$ (extended by $\eta(0) = 0$). Then

$$\sum_{\alpha \in \mathbb{F}_q} \eta(f(\alpha)) = \begin{cases} -\eta(a_2) & \text{if } \Delta \neq 0, \\ (q-1) \cdot \eta(a_2) & \text{if } \Delta = 0. \end{cases}$$

3.9 Jacobsthal Sums

Building further on the theme of character sums twisted by the quadratic character, we now introduce a classical family of sums first studied by Ernst Jacobsthal in 1907 in connection with the representation of primes $p \equiv 1 \bmod 4$ as sums of two squares. Unlike Gaussian and Jacobi sums, which are defined purely multiplicatively, Jacobsthal sums mix an additive shift with the quadratic character η of $\mathbb{F}_q$, which makes them particularly well suited to problems involving quadratic residues and the arithmetic of $\mathbb{F}_q^*$.

Definition 3.9.1: Jacobsthal Sums

For $\alpha \in \mathbb{F}_q^*$ where q is odd and for any $n \in \mathbb{N}$ the Jacobsthal Sum *is defined as*

$$H_n(\alpha) = \sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{n+1} + \alpha \cdot \gamma) = \sum_{\gamma \in \mathbb{F}_q} \eta(\gamma) \cdot \eta(\gamma^n + \alpha)$$

Another associated sum we define is

$$I_n(\alpha) = \sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^n + \alpha)$$

for all $\alpha \in \mathbb{F}_q^*$.

Now for $n = 1$ we get $H_1(\alpha) = 1$ by Theorem 3.8.4 and by Theorem 3.2.2(i), $I_1(\alpha) = 0$ for all $\alpha \in \mathbb{F}_q^*$

The sums H_n and I_n are closely related: in the following theorem we give a recursion relation between I_n and H_m for any $n, m \in \mathbb{N}$.

Lemma 3.9.1

For every $n \in \mathbb{N}$ and every $\alpha \in \mathbb{F}_q^*$,

$$I_{2n}(\alpha) = I_n(\alpha) + H_n(\alpha)$$

Proof: For any $\beta \in \mathbb{F}_q$ let $Z(\beta)$ is the number of $\gamma \in \mathbb{F}_q$ such that $\gamma^2 = \beta$. Hence $Z(\beta) = 1 + \eta(\beta)$. Hence we have

$$\begin{aligned} I_{2n}(\alpha) &= \sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^{2n} + \alpha) \\ &= \sum_{\beta \in \mathbb{F}_q} \eta(\beta^n + \alpha) \cdot Z(\beta) \\ &= \sum_{\beta \in \mathbb{F}_q} \eta(\beta^n + \alpha)(1 + \eta(\beta)) \\ &= \sum_{\beta \in \mathbb{F}_q} \eta(\beta^n + \alpha) + \sum_{\beta \in \mathbb{F}_q} \eta(\beta) \cdot \eta(\beta^n + \alpha) \\ &= I_n(\alpha) + H_n(\alpha) \end{aligned}$$

Hence we have the theorem. ■

3.9.1 Relations between Jacobsthal and Jacobi Sum

Although H_n and I_n are defined using only the quadratic character, they can be evaluated in terms of Jacobi sums for multiplicative characters of higher order. The key observation is that the count of nth-power roots in $\mathbb{F}_q^*$ is a sum of multiplicative characters of order dividing $\gcd(n, q-1)$, and regrouping the definition of I_n accordingly expresses it as a linear combination of Jacobi sums $J(\lambda^j, \eta)$.

Theorem 3.9.2

Let $n \in \mathbb{N}$ and $\alpha \in \mathbb{F}_q^*$, and set $d = \gcd(n, q-1)$. Let $\lambda \in \boldsymbol{\mathcal{M}}_q$ be a multiplicative character of $\mathbb{F}_q$ of order d. Then

$$I_n(\alpha) = \eta(\alpha) \sum_{j=1}^{d-1} \lambda^j(-\alpha) \cdot J(\lambda^j, \eta).$$

Proof: For any $\beta \in \mathbb{F}_q$ let $Z(\beta)$ is the number of $\gamma \in \mathbb{F}_q$ such that $\gamma^n = \beta$. Then we have

$$I_n(\alpha) = \sum_{\gamma \in \mathbb{F}_q} \eta(\gamma^n + \alpha) = \sum_{\beta \in \mathbb{F}_q} \eta(\beta + \alpha) \cdot Z(\beta)$$

Suppose $\beta \neq 0$. Then by Theorem 3.2.5(iii) we have

$$Z(\beta) = \frac{1}{q-1} \sum_{\gamma \in \mathbb{F}_q^*} \sum_{\psi \in \boldsymbol{\mathcal{M}}_q} \psi(\gamma^n)\overline{\psi}(\beta) = \frac{1}{q-1} \sum_{\psi \in \boldsymbol{\mathcal{M}}_q} \overline{\psi}(\beta) \sum_{\gamma \in \mathbb{F}_q^*} \psi^n(\gamma)$$

Now by Theorem 3.2.5(i), ψ^n is trivial then $\sum_{\gamma \in \mathbb{F}_q^*} \psi^n(\gamma) = q-1$ and otherwise its 0. Now ψ^n is trivial if and only if $\psi = \overline{\lambda}^j$ for all $j \in \{0, 1, \ldots, d-1\}$. Then we have

$$Z(\beta) = \sum_{j=0}^{d-1} \overline{\lambda}^j(\beta)$$

Suppose $\beta = 0$ then $Z(\beta) = 1$ as 0 is the only solution. Therefore this also holds for $\beta = 0$. Hence we have

$$\begin{aligned}
I_n(\alpha) &= \sum_{\beta\in\mathbb{F}_q} \eta(\beta+\alpha) \sum_{j=0}^{d-1} \lambda^j(\beta) \\
&= \sum_{j=0}^{d-1} \sum_{\beta\in\mathbb{F}_q} \eta(\beta+\alpha)\lambda^j(\beta) \\
&= \eta(-1) \sum_{j=0}^{d-1} \sum_{\beta\in\mathbb{F}_q} \eta(-(\beta+\alpha))\lambda^j(\beta) \\
&= \eta(-1) \sum_{j=0}^{d-1} J_{-\alpha}(\lambda^j, \eta) \\
&= \eta(-1) \sum_{j=0}^{d-1} (\lambda^j \cdot \eta)(-\alpha) J(\lambda^j, \eta) \\
&= \eta(\alpha) \sum_{j=0}^{d-1} \lambda^j(-\alpha) J(\lambda^j, \eta)
\end{aligned}$$

Hence we have the theorem. ■

The Jacobsthal sum H_n satisfies a similar Jacobi-sum representation, but now involving a character λ of order $2d$ rather than d, and with the parity of the power of 2 dividing $q-1$ relative to n playing a decisive role: when all the 2-power content of $q-1$ already divides n, the sum H_n vanishes identically.

Lemma 3.9.3

Let $n \in \mathbb{N}$ and $\alpha \in \mathbb{F}_q^*$, and set $d = \gcd(n, q-1)$. If the largest power of 2 dividing $q-1$ also divides n, then $H_n(\alpha) = 0$. Otherwise, let $\lambda \in \mathcal{M}_q$ be a multiplicative character of $\mathbb{F}_q$ of order $2d$; then

$$H_n(\alpha) = \eta(\alpha) \cdot \lambda(-1) \sum_{j=0}^{d-1} \lambda^{2j+1}(\alpha) \cdot J(\lambda^{2j+1}, \eta).$$

Proof: Since the largest power of 2 dividing $q-1$ also divides n we have $d = \gcd(n, q-1) = \gcd(2n, q-1)$. Therefore by Theorem 3.9.2, $I_n(\alpha) = I_{2n}(\alpha)$. Therefore by Lemma 3.9.1 we get $H_n(\alpha) = 0$.

Now suppose that is not the case. Then $\gcd(2n, q-1) = 2d$. Then by Theorem 3.9.2 we get

$$I_{2n}(\alpha) = \eta(\alpha) \sum_{j=1}^{2d-1} \lambda^j(-\alpha) J(\lambda^j, \eta) \tag{3.18}$$

Since $\operatorname{ord}(\lambda^2) = d$ using the same theorem for $I_n(\alpha)$ and λ^2 we get

$$I_n(\alpha) = \eta(\alpha) \sum_{j=0}^{d-1} \lambda^{2j}(-\alpha) J(\lambda^{2j}, \eta) \tag{3.19}$$

Therefore from Lemma 3.9.1 we have $H_n(\alpha) = I_{2n}(\alpha) - I_n(\alpha)$. So subtracting (3.19) from (3.18) we get the expression. ■

Therefore by Theorem 3.4.6 we get for all $\alpha \in \mathbb{F}_q^*$, $|I_n(\alpha)| \le (d-1)q^{1/2}$.

3.9.2 Yet Another Proof of Fermat's Two Square Theorem

We close this section with the classical application of Jacobsthal sum. Using H_2 one can compute the two integers A, B in the representation $p = A^2 + B^2$ (guaranteed by Theorem 3.4.11) directly, and moreover with a canonical sign choice determined by a congruence modulo 4. Here $p \equiv 1 \bmod 4$ is prime, η is the quadratic character of $\mathbb{F}_p$, and $\lambda \in \boldsymbol{\mathcal{M}}_p$ is a multiplicative character of $\mathbb{F}_p$ of order 4, so that $\lambda^2 = \eta$ and $\lambda^3 = \overline{\lambda}$.

By Theorem 3.9.3 applied with $n = 2$ and hence $d = \gcd(2, p-1) = 2$,

$$H_2(1) = \lambda(-1) \sum_{j=0}^{1} \lambda^{2j+1}(1) \cdot J(\lambda^{2j+1}, \eta) = \lambda(-1) \left(J(\lambda, \eta) + J(\lambda^3, \eta) \right).$$

Since $\lambda^3 = \overline{\lambda}$ and η is real, $J(\lambda^3, \eta) = \overline{J(\lambda, \eta)}$, so $H_2(1) = 2\lambda(-1) \cdot \Re J(\lambda, \eta)$. In the notation of Theorem 3.4.11, $J(\lambda, \eta) = A + iB$ with $A, B \in \mathbb{Z}$ and $p = A^2 + B^2$, so

$$\Re J(\lambda, \eta) = A = \tfrac{1}{2}\lambda(-1) \cdot H_2(1).$$

A similar computation starting from an $\alpha \in \mathbb{F}_p$ with $\eta(\alpha) = -1$ uses $\lambda(\alpha) = \pm i$ and $\lambda^3(\alpha) = -\lambda(\alpha)$ to give

$$H_2(\alpha) = \eta(\alpha)\lambda(-1) \left(\lambda(\alpha) J(\lambda, \eta) - \lambda(\alpha) \overline{J(\lambda, \eta)} \right) = -\lambda(-1)\lambda(\alpha) \cdot 2i \cdot \Im J(\lambda, \eta),$$

which identifies $\Im J(\lambda, \eta) = B$ up to the unit factor $-\lambda(-1)\lambda(\alpha) \in \{\pm 1, \pm i\}$; absorbing this factor gives $B = \frac{1}{2} H_2(\alpha)$ (up to sign).

To pin down the sign of A, we expand $H_2(1)$ directly. Pairing γ with $-\gamma$ and using $\eta(-1) = 1$ (since $p \equiv 1 \bmod 4$) gives

$$H_2(1) = \sum_{\gamma=1}^{p-1} \eta(\gamma) \cdot \eta(\gamma^2 + 1) = 2 \sum_{\gamma=1}^{(p-1)/2} \eta(\gamma) \cdot \eta(\gamma^2 + 1),$$

so $\frac{1}{2} H_2(1) = \sum_{\gamma=1}^{(p-1)/2} \eta(\gamma)\eta(\gamma^2 + 1)$. On the other hand Theorem 3.8.4 applied to $f(X) = X^2 + 1$ (discriminant $-4 \neq 0$ and $\eta(1) = 1$) gives $\sum_{\gamma \in \mathbb{F}_p} \eta(\gamma^2 + 1) = -1$, so $\sum_{\gamma=1}^{(p-1)/2} \eta(\gamma^2 + 1) = -1$ (excluding the zero term). Subtracting and reducing modulo 4, using the fact that $(\eta(\gamma) - 1)(\eta(\gamma^2 + 1) - 1) \equiv 0 \bmod 4$ whenever $\eta(\gamma^2 + 1) \neq 0$, one obtains after a short computation

$$\tfrac{1}{2} H_2(1) + 1 \equiv \tfrac{3-p}{2} - \lambda(-1) \bmod 4.$$

Since $\lambda(-1) = 1$ when $p \equiv 1 \bmod 8$ and $\lambda(-1) = -1$ when $p \equiv 5 \bmod 8$, in both cases $\frac{1}{2} H_2(1) + 1 \equiv 0 \bmod 4$, i.e. $\frac{1}{2} H_2(1) \equiv -1 \bmod 4$.

Hence setting $A = \frac{1}{2}\lambda(-1) H_2(1)$ and $B = \frac{1}{2} H_2(\alpha)$ (for any $\alpha \in \mathbb{F}_p$ with $\eta(\alpha) = -1$), we obtain integers with $A^2 + B^2 = p$, A odd, B even, and the normalization $A \equiv -1 \bmod 4$. With this normalization A is uniquely determined: if $p = A^2 + B^2 = C^2 + D^2$ with A, C odd and $A \equiv C \equiv -1 \bmod 4$, then choosing $h, k \in \mathbb{Z}$ with $A \equiv hB$, $C \equiv kD \bmod p$ gives $h^2 + 1 \equiv k^2 + 1 \equiv 0 \bmod p$, hence $C \equiv \pm hD \bmod p$; writing $C_1 \equiv hD \bmod p$ and $C = \pm C_1$, we compute

$$p^2 = (A^2 + B^2)(C_1^2 + D^2) = (AC_1 + BD)^2 + (AD - BC_1)^2,$$

and both $AC_1 + BD \equiv (h^2 + 1)BD \equiv 0 \bmod p$ and $AD - BC_1 \equiv 0 \bmod p$. Dividing by p^2 gives $1 = (\pm 1)^2 + 0^2$, forcing $AD - BC_1 = 0$ and $|AC_1 + BD| = p$. Since $\gcd(A, B) = \gcd(C_1, D) = 1$, it follows that $A = \pm C_1$ and hence $A = \pm C$; the congruence $A \equiv C \equiv -1 \bmod 4$ then forces $A = C$.

Thus the Jacobsthal sum $\frac{1}{2}\lambda(-1) H_2(1)$ produces the canonical integer A in Fermat's representation $p = A^2 + B^2$, and $\frac{1}{2} H_2(\alpha)$ produces B up to sign.

3.10 Salié Sums

The Kloosterman sums of the previous section are notoriously difficult to evaluate in closed form – even their size is controlled only by the deep theorem of Weil. Remarkably, a very mild modification of the Kloosterman sum, obtained by inserting the quadratic character η as a weight, produces a family of sums for which exact evaluations *are* possible. These are the *Salié sums*, introduced by Hans Salié in 1931.

Definition 3.10.1: Salié Sum

Let $\mathbb{F}_q$ be a finite field with q odd, and let $\chi, \tau \in \boldsymbol{\chi}_q$ be non-trivial additive characters of $\mathbb{F}_q$. Then the Salié sum *associated to χ and τ is defined as*

$$S(\chi, \tau) = \sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma) \cdot \chi(\gamma) \cdot \tau(\gamma^{-1})$$

Theorem 3.10.1

Let $\mathbb{F}_q$ be a finite elements with q being odd and $\chi, \tau \in \boldsymbol{\chi}_q$ be non-trivial additive character of $\mathbb{F}_q$. Let $a \in \mathbb{F}_q$ such that for all $\gamma \in \mathbb{F}_q$, $\tau(\gamma) = \chi(a \cdot \gamma)$. Then we have

$$S(\chi, \tau) = G(\eta, \chi) \sum_{b\colon b^2 = 4a} \chi(b)$$

and

$$|S(\chi, \tau)| \leq 2 \cdot q^{1/2}$$

Proof: Since $\tau(\gamma) = \chi(a \cdot \gamma)$ for all $\gamma \in \mathbb{F}_q$ we have $\chi(\gamma) \cdot \tau(\gamma^{-1}) = \chi(\gamma) \cdot \chi(a \cdot \gamma^{-1}) = \chi(\gamma + a\gamma^{-1})$, so $S(\chi, \tau) = \sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\chi(\gamma + a\gamma^{-1})$.

Step 1: Multiplicative Fourier expansion. For $b \in \mathbb{F}_q^*$ let $\chi_b \in \boldsymbol{\chi}_q$ denote the additive character $\chi_b(\gamma) = \chi(b\gamma)$, similar to as given by Theorem 3.2.1. Define $\varphi : \mathbb{F}_q^* \to \mathbb{C}$ by

$$\varphi(b) = S(\chi_b, \tau) = \sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\chi(b\gamma + a\gamma^{-1}).$$

By Theorem 3.2.5, the multiplicative characters form an orthonormal basis for functions on $\mathbb{F}_q^*$, so we may expand

$$\varphi(b) = \sum_{\lambda \in \boldsymbol{M}_q} \hat{\varphi}(\lambda)\lambda(b), \quad \text{where} \quad \hat{\varphi}(\lambda) = \frac{1}{q-1} \sum_{b \in \mathbb{F}_q^*} \varphi(b)\overline{\lambda}(b).$$

Step 2: Computing the Fourier coefficients. Substituting the definition of φ and interchanging the order of summation:

$$\begin{aligned}\hat{\varphi}(\lambda) &= \frac{1}{q-1} \sum_{b \in \mathbb{F}_q^*} \overline{\lambda}(b) \sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\chi(b\gamma + a\gamma^{-1}) \\ &= \frac{1}{q-1} \sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\chi(a\gamma^{-1}) \sum_{b \in \mathbb{F}_q^*} \overline{\lambda}(b)\chi(b\gamma).\end{aligned}$$

Substituting $b \mapsto b\gamma^{-1}$ in the inner sum gives $\sum_{b \in \mathbb{F}_q^*} \overline{\lambda}(b)\chi(b\gamma) = \lambda(\gamma) \sum_{b \in \mathbb{F}_q^*} \overline{\lambda}(b)\chi(b) = \lambda(\gamma) \cdot G(\overline{\lambda}, \chi)$. Hence

$$\hat{\varphi}(\lambda) = \frac{G(\overline{\lambda}, \chi)}{q-1} \sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\lambda(\gamma)\chi(a\gamma^{-1}).$$

Substituting $\gamma \mapsto \gamma^{-1}$ in the remaining sum:

$$\sum_{\gamma \in \mathbb{F}_q^*} \eta(\gamma)\lambda(\gamma)\chi(a\gamma^{-1}) = \sum_{\gamma \in \mathbb{F}_q^*} \overline{\eta}(\gamma)\overline{\lambda}(\gamma)\chi(a\gamma) = G(\overline{\eta} \cdot \overline{\lambda}, \chi_a)$$

where $\chi_a(\gamma) = \chi(a\gamma)$. Since η is real-valued, $\overline{\eta} = \eta$. By Lemma 3.3.2(i), $G(\lambda, \chi_a) = \overline{\lambda}(a)G(\lambda, \chi)$, so

$$G(\overline{\eta} \cdot \overline{\lambda}, \chi_a) = \eta(a)\lambda(a) \cdot G(\overline{\eta} \cdot \overline{\lambda}, \chi).$$

Altogether:

$$\hat{\varphi}(\lambda) = \frac{\eta(a)\lambda(a) \cdot G(\overline{\lambda}, \chi) \cdot G(\overline{\eta} \cdot \overline{\lambda}, \chi)}{q-1}. \tag{3.20}$$

Step 3: Applying the Hasse-Davenport product formula. By Corollary 3.4.12 applied with $m = 2$, the quadratic character η, and $\overline{\lambda}$ in place of ψ, we have

$$G(\overline{\lambda}, \chi) \cdot G(\overline{\lambda}\eta, \chi) = \lambda(4) \cdot G(\overline{\lambda}^2, \chi) \cdot G(\eta, \chi).$$

Since $\overline{\eta} \cdot \overline{\lambda} = \overline{\lambda}\eta$, substituting into (3.20):

$$\hat{\varphi}(\lambda) = \frac{\eta(a)G(\eta, \chi)}{q-1} \cdot \lambda(a) \cdot \lambda(4) \cdot G(\overline{\lambda}^2, \chi) = \frac{\eta(a)G(\eta, \chi)}{q-1} \cdot \lambda(4a) \cdot G(\overline{\lambda}^2, \chi).$$

Step 4: Final Step. Since $\lambda(1) = 1$ for every $\lambda \in \boldsymbol{\mathcal{M}}_q$:

$$\begin{aligned} S(\chi, \tau) = \varphi(1) = \sum_{\lambda \in \boldsymbol{\mathcal{M}}_q} \hat{\varphi}(\lambda) &= \frac{\eta(a)G(\eta, \chi)}{q-1} \sum_{\lambda \in \boldsymbol{\mathcal{M}}_q} \lambda(4a) \cdot G(\overline{\lambda}^2, \chi) \\ &= \frac{\eta(a)G(\eta, \chi)}{q-1} \sum_{\lambda \in \boldsymbol{\mathcal{M}}_q} \lambda(4a) \sum_{\gamma \in \mathbb{F}_q^*} \overline{\lambda}^2(\gamma)\chi(\gamma) \\ &= \frac{\eta(a)G(\eta, \chi)}{q-1} \sum_{\gamma \in \mathbb{F}_q^*} \chi(\gamma) \sum_{\lambda \in \boldsymbol{\mathcal{M}}_q} \lambda(4a\gamma^{-2}) \end{aligned}$$

By the orthogonality of multiplicative characters (Theorem 3.2.5(ii)):

$$\sum_{\lambda \in \boldsymbol{\mathcal{M}}_q} \lambda(4a\gamma^{-2}) = \begin{cases} q-1 & \text{if } \gamma^2 = 4a, \\ 0 & \text{otherwise.} \end{cases}$$

Hence $S(\chi, \tau) = \eta(a) \cdot G(\eta, \chi) \sum_{b:b^2=4a} \chi(b)$. If $\eta(a) = -1$ then $4a$ is a non-square so $b^2 = 4a$ has no solution and the sum is 0; if $\eta(a) = 1$ the factor is 1. In both cases:

$$S(\chi, \tau) = G(\eta, \chi) \sum_{b:\, b^2=4a} \chi(b).$$

Therefore we have the theorem. ■

Lemma 3.10.2

Let $\mathbb{F}_q$ be a finite elements with q being odd and $\chi, \tau \in \boldsymbol{\mathcal{X}}_q$ be non-trivial additive character of $\mathbb{F}_q$. Let $a \in \mathbb{F}_q$ such that for all $\gamma \in \mathbb{F}_q$, $\tau(\gamma) = \chi(a \cdot \gamma)$. Then $|S(\chi, \tau)| \leq 2 \cdot q^{1/2}$.

Proof: Since $a \neq 0$, the equation $b^2 = 4a$ has either 0 or 2 solutions in $\mathbb{F}_q$. From Theorem 3.10.1 if 0 solutions, $S(\chi, \tau) = 0$ and if $\pm b_0$ are the two solutions, $S(\chi, \tau) = G(\eta, \chi)\,(\chi(b_0) + \chi(-b_0))$. Each term $G(\eta, \chi)\chi(\pm b_0)$ is a product of the Gauss sum with a root of unity. By Theorem 3.3.1, $|G(\eta, \chi)| = q^{1/2}$, so $|S(\chi, \tau)| \leq 2q^{1/2}$. ■

Chapter 4

Stepanov and Bombieri's Method

Chapter 5

Equations over Finite Fields

5.1 Special L-functions

We write alternate proof of Davenport-Hasse Theroem from Schmidt 2.6,7,8,9,10

Chapter 6

Pseudopolynomials

Chapter 7

Decoding from Quadratic Character Evaluations

Chapter 8

Bibliography

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Chapter A

Appendix

A.1 Fermat's Little Theorem

Fermat's Little Theorem is one of the oldest results in elementary number theory, recorded in a 1640 letter from Pierre de Fermat to Frénicle de Bessy. It underlies essentially every computation we perform in the multiplicative group $\mathbb{F}_p^*$ – from the very existence of the canonical additive character of $\mathbb{F}_q$ (where $q = p^r$), through the character congruence $\psi(l) \equiv l^f \pmod{P}$ used in the arithmetic proof of the Davenport–Hasse relation, down to the mod-q reduction of Gauss sums appearing in the proof of the Law of Quadratic Reciprocity. We record it here for completeness.

Theorem A.1.1 Fermat's Little Theorem

Let p be a prime. Then for every integer a we have

$$a^p \equiv a \pmod{p},$$

and in particular, if $p \nmid a$ then $a^{p-1} \equiv 1 \pmod{p}$.

Proof: The non-zero residues modulo p form a group $\mathbb{F}_p^* = \{1, 2, \ldots, p-1\}$ under multiplication of order $p-1$. By Lagrange's theorem every element of a finite group satisfies $g^{|G|} = e$, so $a^{p-1} \equiv 1 \pmod{p}$ for every $a \in \mathbb{F}_p^*$, i.e. every a with $p \nmid a$. Multiplying both sides by a gives $a^p \equiv a \pmod{p}$ for such a; and the congruence $a^p \equiv a \pmod{p}$ is trivially true when $p \mid a$, so it holds for every integer a. ∎

A.2 Waring's Formula

Waring's Formula expresses each power-sum symmetric polynomial $s_k = x_1^k + x_2^k + \cdots + x_n^k$ as an explicit polynomial in the elementary symmetric polynomials $\sigma_1, \ldots, \sigma_n$. It is the source of the closed-form reduction of lifted Kloosterman sums $K(\chi^{(s)}; a, b)$ to the ground-field sum $K(\chi; a, b)$ in Theorem 3.7.3: applying Waring's formula to $\omega_1^s + \omega_2^s$ with $\sigma_1 = \omega_1 + \omega_2 = -K(\chi; a, b)$ and $\sigma_2 = \omega_1\omega_2 = q$ produces the binomial-coefficient expansion. We state the general identity here for reference.

Theorem A.2.1 Waring's Formula

Let $x_1, \ldots, x_n$ be indeterminates, let

$$\sigma_j = \sum_{1 \le i_1 < i_2 < \cdots < i_j \le n} x_{i_1} x_{i_2} \cdots x_{i_j} \qquad (j = 1, \ldots, n)$$

be the elementary symmetric polynomials in $x_1, \ldots, x_n$, and let

$$s_k = x_1^k + x_2^k + \cdots + x_n^k$$

be the kth power-sum. Then for every integer $k \ge 1$,

$$s_k = \sum (-1)^{i_2 + i_4 + i_6 + \cdots} \cdot \frac{(i_1 + i_2 + \cdots + i_n - 1)! \cdot k}{i_1! \cdot i_2! \cdots i_n!} \cdot \sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n},$$

where the summation runs over all n-tuples $(i_1, \ldots, i_n)$ of non-negative integers satisfying $i_1 + 2i_2 + \cdots + ni_n = k$. Each coefficient of $\sigma_1^{i_1} \sigma_2^{i_2} \cdots \sigma_n^{i_n}$ is an integer.